Hypothalamic & Pituitary Hormones

Block 7/Week 1

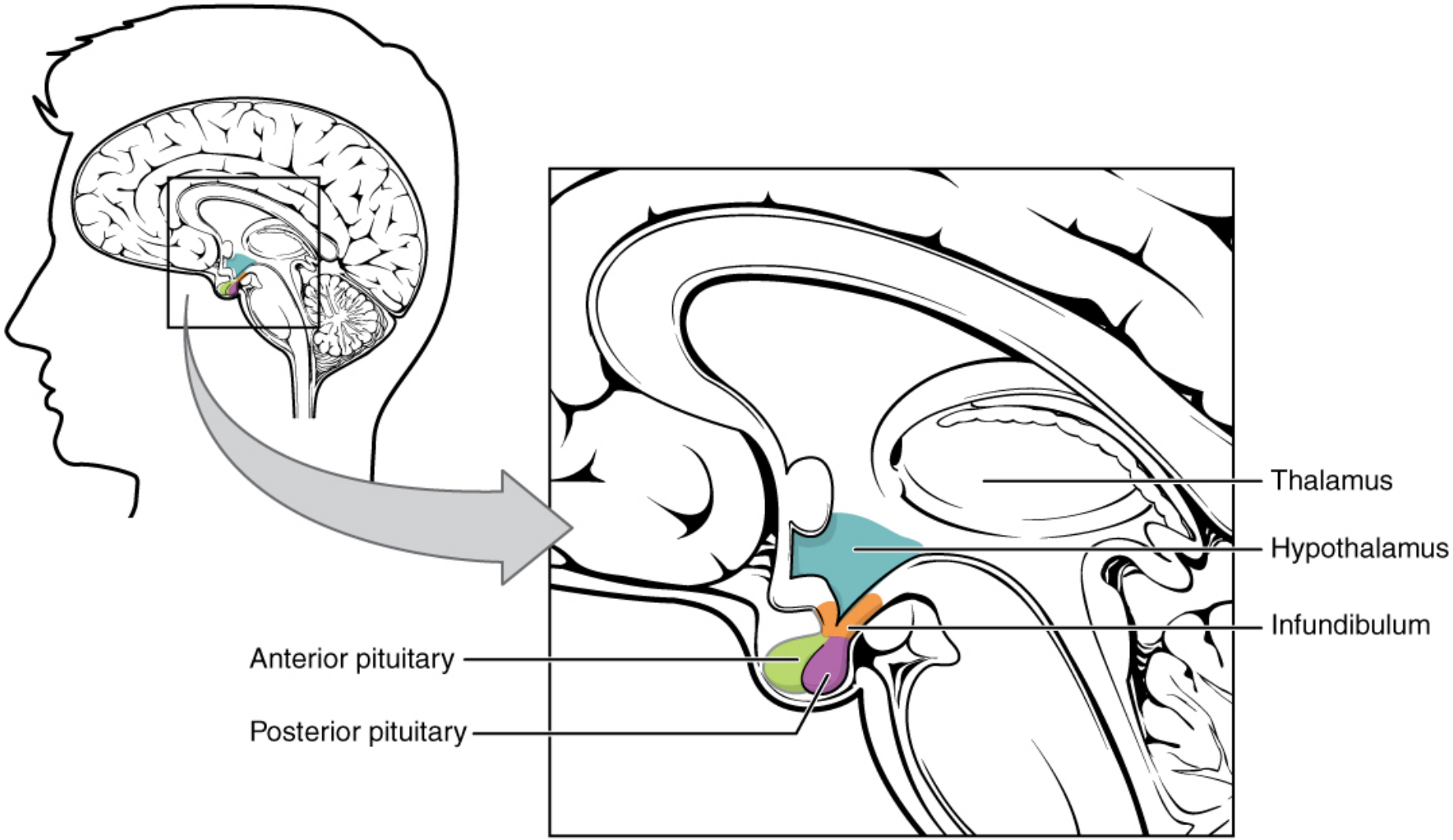
UIC UNIVERSITY OF ILLINOIS
COLLEGE OF MEDICINE

Developed by C. Stocco with contributions from J. Regan and H. Clements-Jewery

Learning Objectives

1. Describe the anterior pituitary lobe with respect to histology, cell types, vascular supply, development, and anatomical function relative to the hypothalamus.
2. Distinguish between releasing and inhibitory factors from hypothalamic neurons and trophic hormones produced by cells of the adenohypophysis, to include their route of transport from the hypothalamus to the anterior pituitary.
3. Describe the five endocrine cell types within the anterior pituitary, their primary hypothalamic regulation, the tropic hormone(s) secreted, and target endocrine gland(s).
4. Describe negative feedback control of anterior pituitary hormone secretion at the hypothalamic and pituitary levels.
5. Describe the posterior pituitary with respect to cell types, vascular supply, development, and structural connections to the hypothalamus.
6. List the target organs, stimuli and functional effects of oxytocin.
7. Describe the stimuli and mechanisms that control vasopressin secretion.
8. List the target cells for vasopressin and explain why vasopressin is also known as antidiuretic hormone.
9. Identify primary, secondary and tertiary endocrine disease based on plasma levels of hypothalamic-releasing hormones, stimulating hormones, and target hormones.

The pituitary gland (hypophysis) is a small endocrine structure at the base of the forebrain



Pituitary Gland (Hypophysis)

Adenohypophysis/Anterior lobe

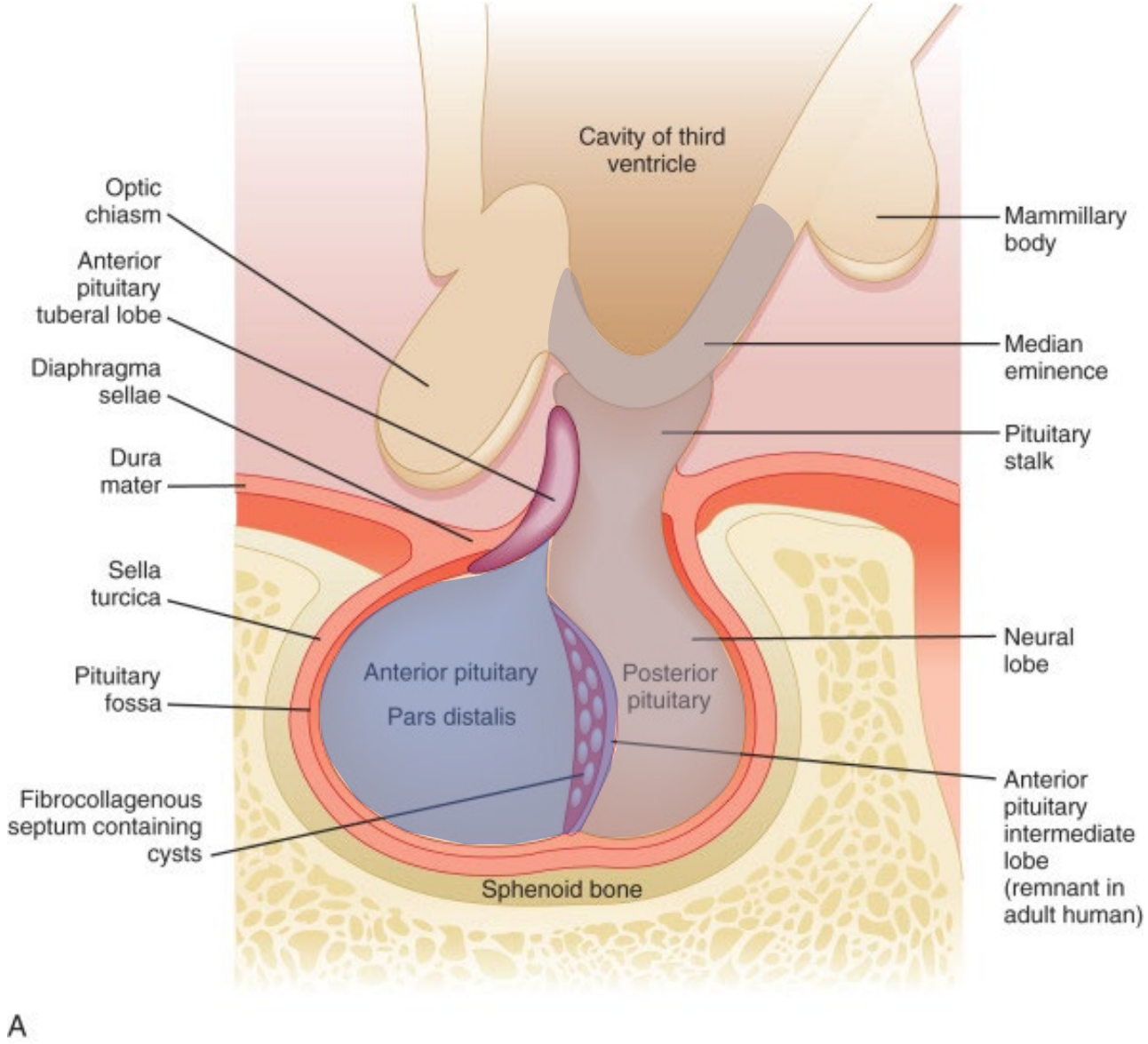
Epithelial tissue

- Pars distalis (90% of lobe)
- pars tuberalis (wraps around infundibulum)

Neurohypophysis

Neural tissue

- Posterior lobe/pars nervosa
- Infundibulum (pituitary stalk)
- Median eminence

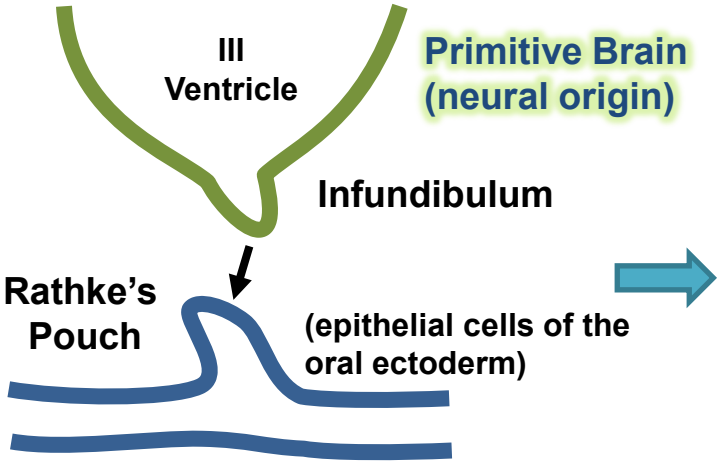


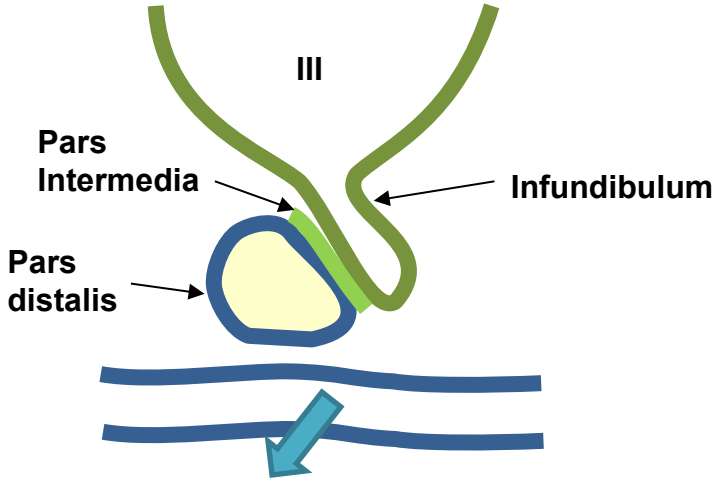
Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 41, 733-752

The anterior and posterior pituitary originate from separate embryological structures

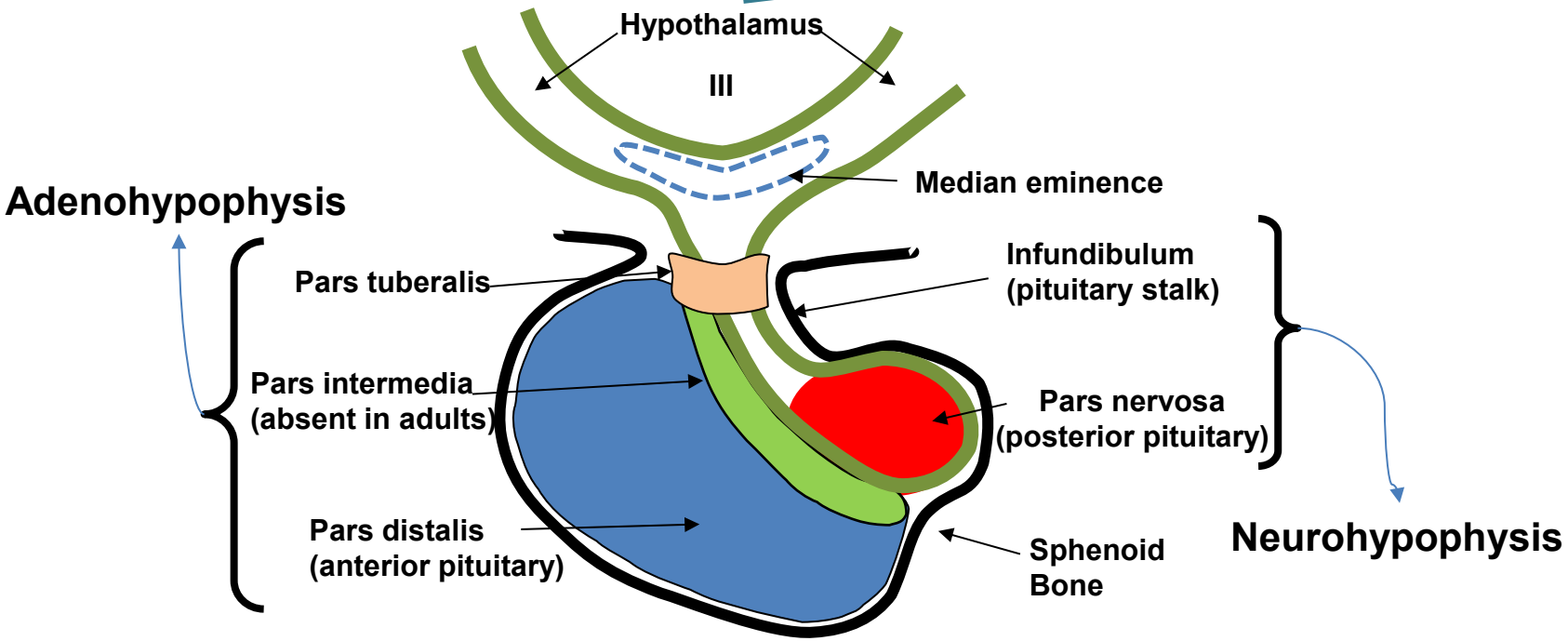
LO1,5

Anterior pituitary originates from invagination of pharyngeal epithelium





Posterior pituitary originates from neural tissue outgrowth from the hypothalamus

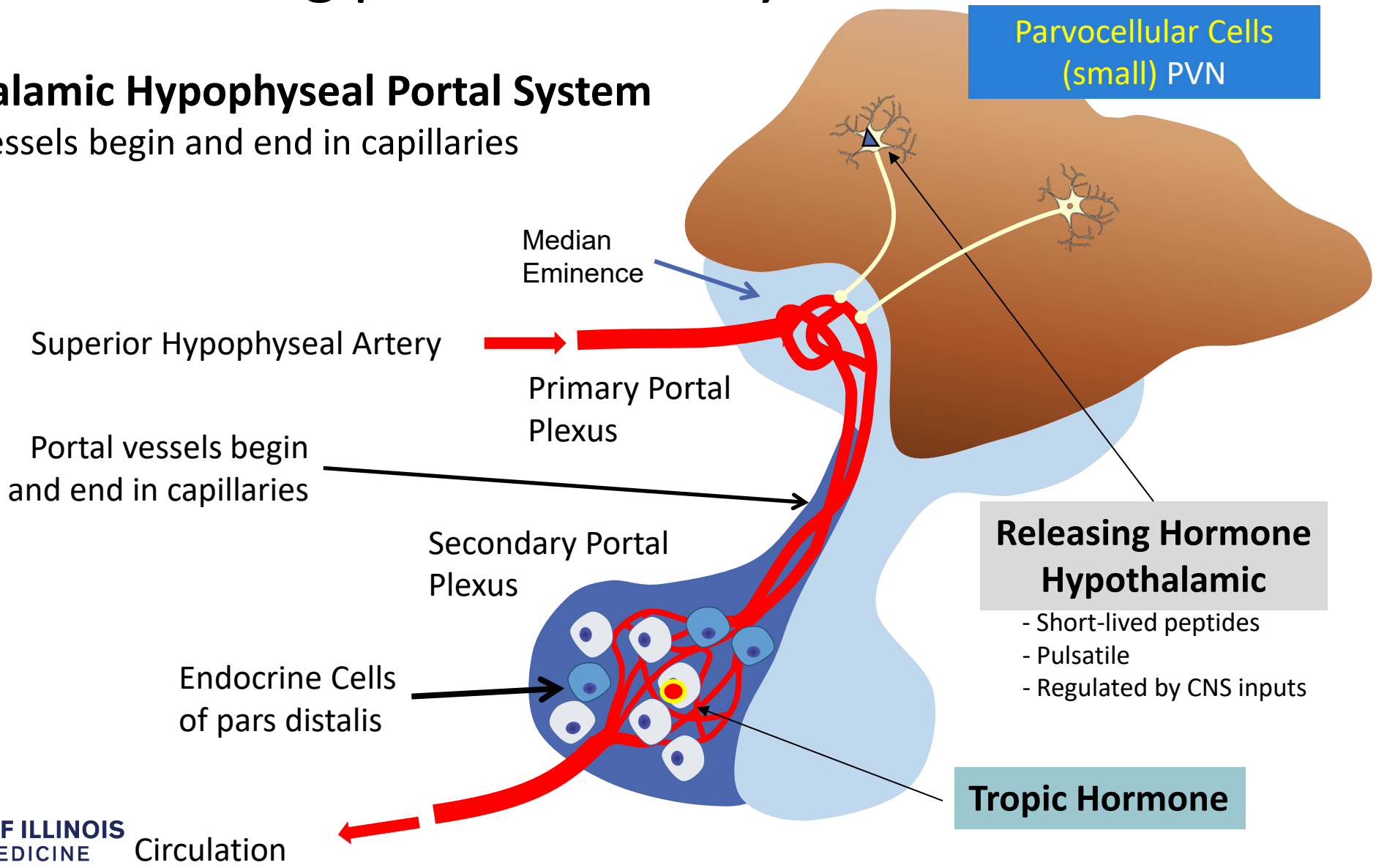


Adenohypophysis

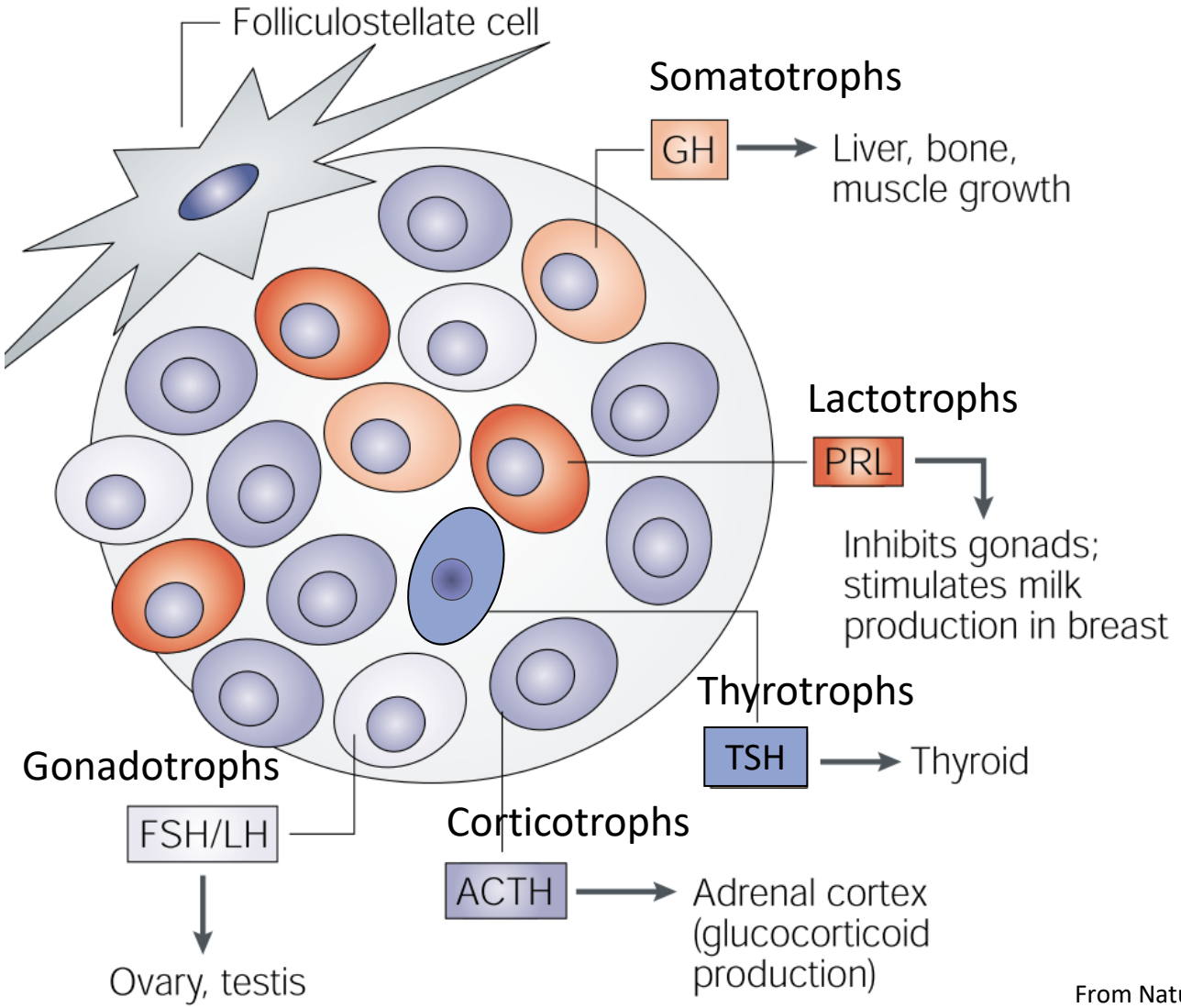
Hypothalamus and adenohypophysis communicate via vasculature using portal vessel system

Hypothalamic Hypophyseal Portal System

- Portal vessels begin and end in capillaries



The anterior pituitary contains 5 types of endocrine cells, from which 6 hormones are secreted



From Nature Reviews Cancer 2:836

Summary of adenohypophysis cell types:

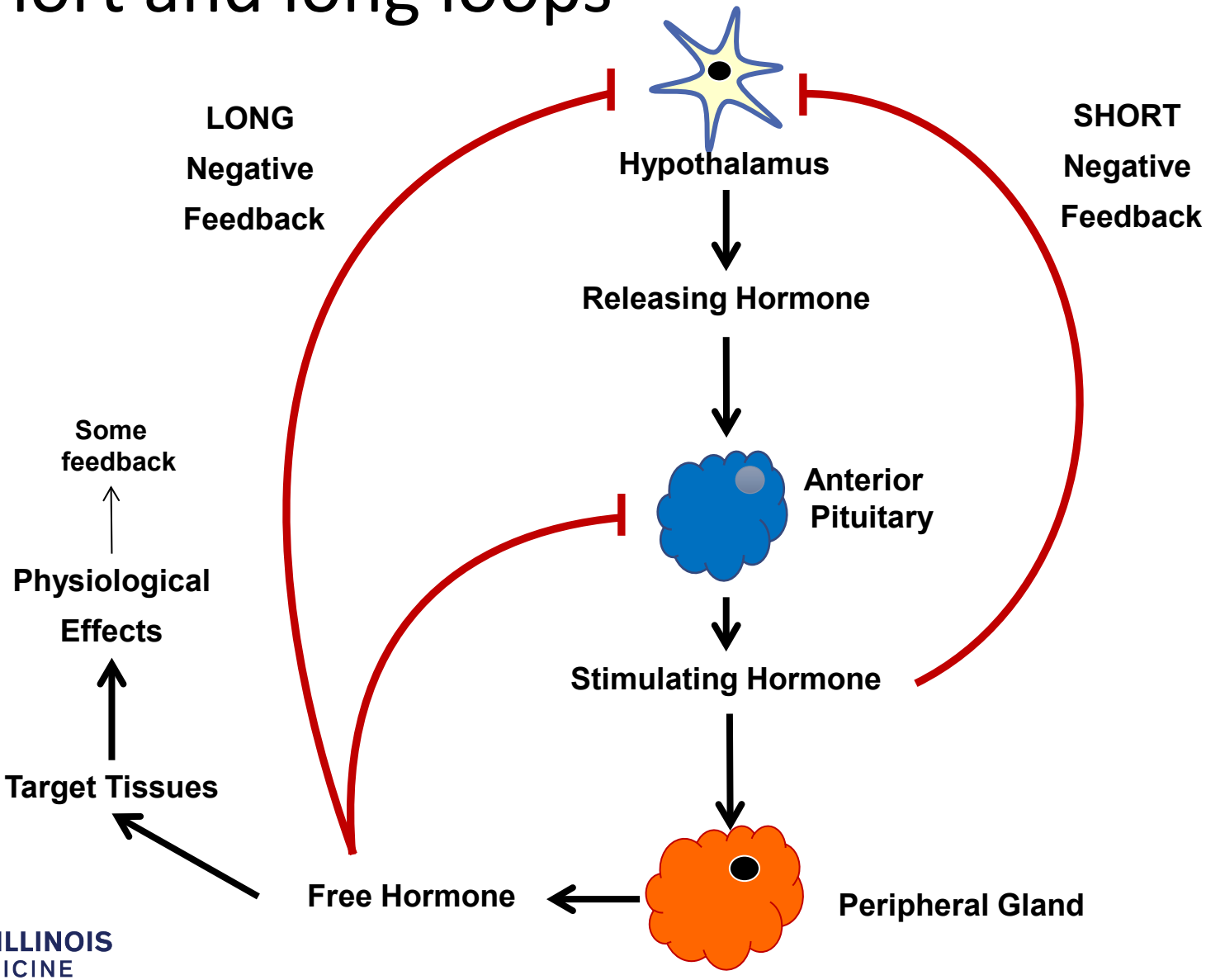
Hormonal production and action, hypothalamic regulation, and feedback regulation

	Corticotrope	Thyrotrope	Gonadotrope	Somatotrope	Lactotrope
Primary hypothalamic regulation	Corticotropin-releasing hormone (CRH)- stimulatory	Thyrotropin-releasing hormone (TRH)- stimulatory	Gonadotropin-releasing hormone (GnRH)- stimulatory	Growth hormone-releasing hormone (GHRH)- stimulatory Somatostatin- inhibitory	Dopamine- inhibitory
Tropic hormone secreted	Adrenocorticotropic hormone (ACTH)- protein	Thyroid-stimulating hormone (TSH)- glycoprotein	Follicle-stimulating hormone (FSH) and luteinizing hormone (LH)- glycoproteins	Growth hormone (GH)- protein	Prolactin (PRL)- protein
Target endocrine gland	Zona fasciculate and zona reticularis of adrenal cortex	Thyroid epithelium	Ovary (theca and granulosa) Testis (Leydig and Sertoli cells)	Liver, and also direct actions especially metabolic effects	No target endocrine gland, not part of an endocrine axis
Peripheral hormone involved in negative feedback	Cortisol	Triiodothyronine (T3)	Estrogen[a], progesterone, testosterone, inhibin[b]	IGF-1, GH (short loop)	None

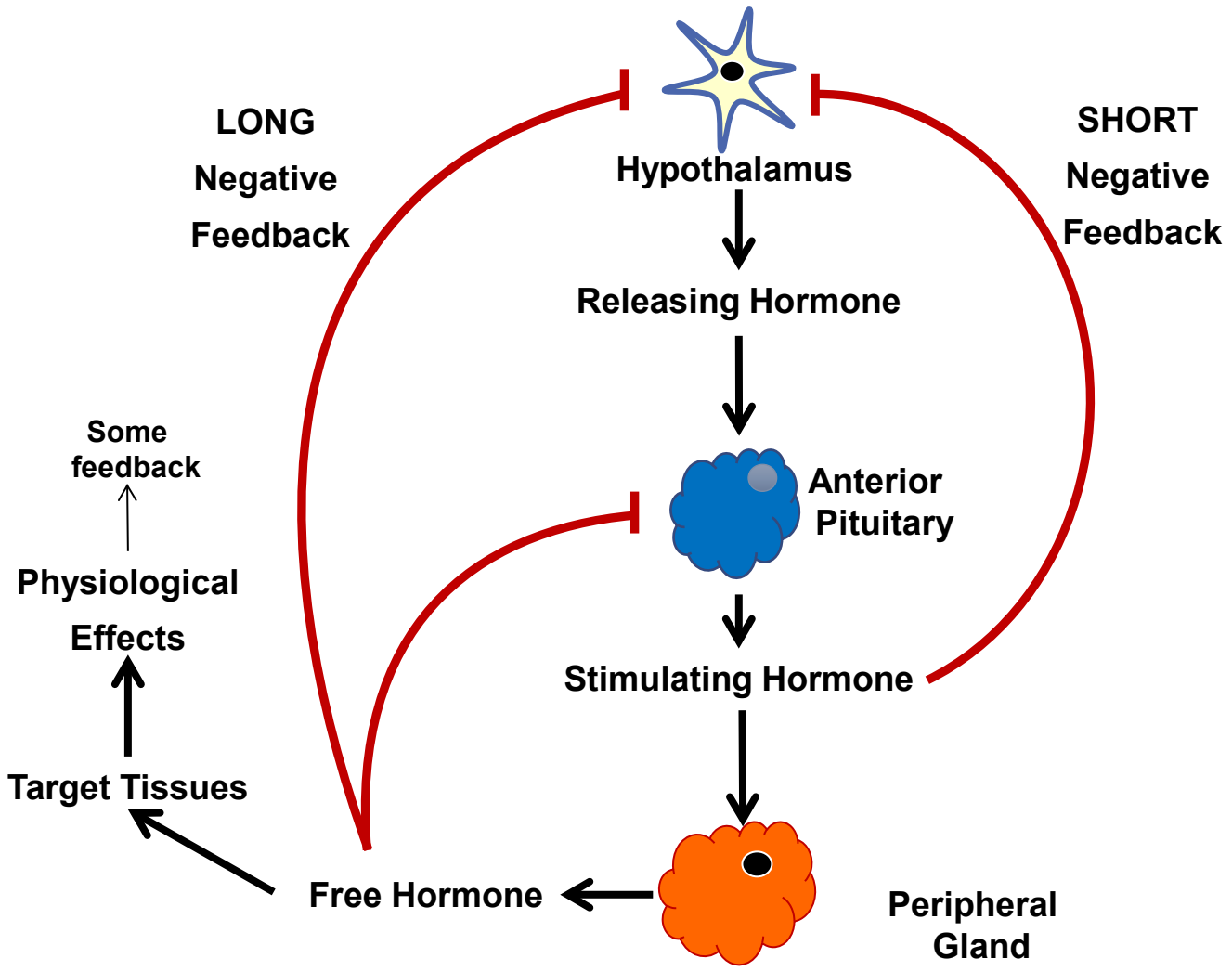
[a] estrogen can also have a positive feedback in women
[b] inhibin selectively inhibits release of FSH from the gonadotropes

From Table 41.1, Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 41, 733-752

Negative feedback regulation via endocrine axis occurs by both short and long loops



Activity of an axis is normally maintained at a **set point**



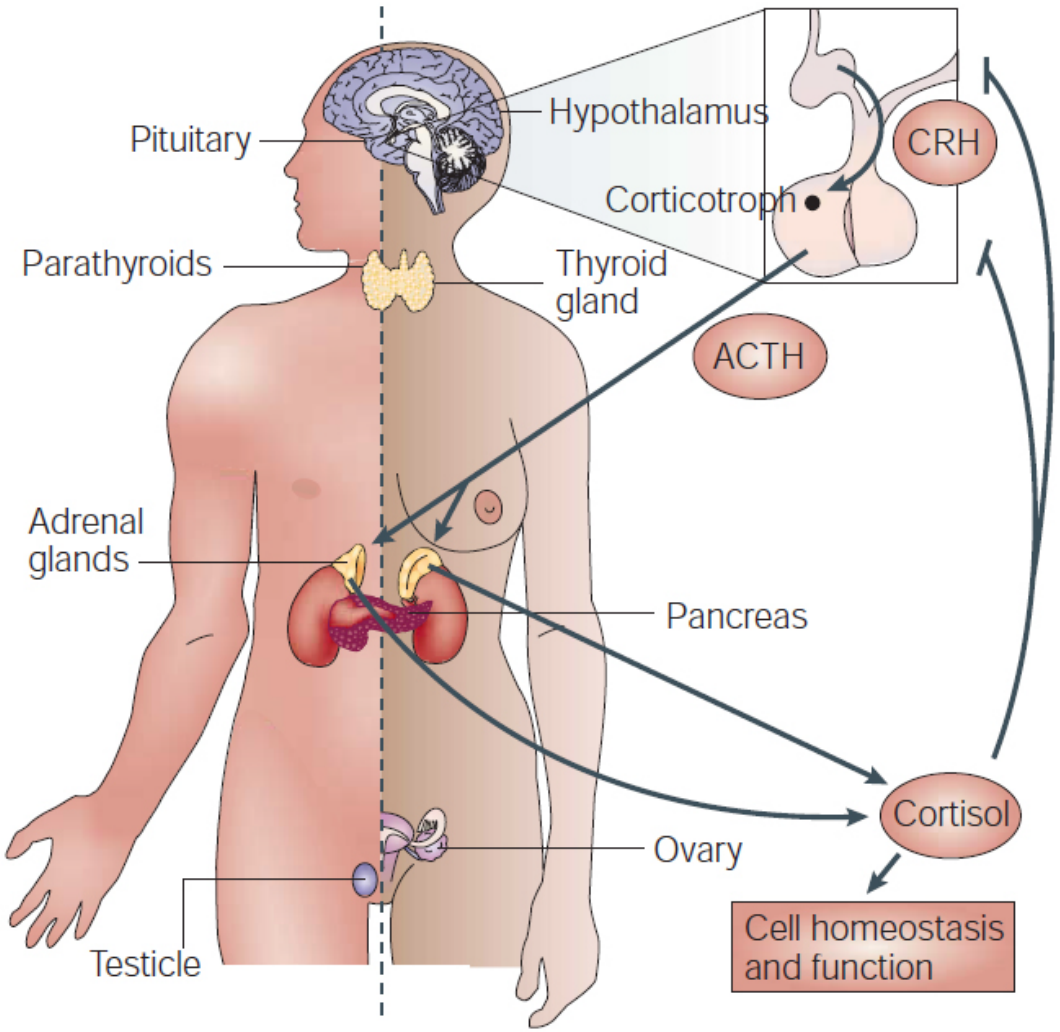
Set point:

- Varies by individual, usually within a normal range
- Determined by integration of hypothalamic stimulation and peripheral hormone negative feedback

CNS input to hypothalamus:

- regulates hypothalamic neuron secretions (pulsatile or rhythmic secretion)
- integrates multiple sources of information to set or change the set point of an axis
- Broad range of complex neurogenic states can alter pituitary function

Hypothalamus-Pituitary-Adrenal Axis



- Secretion of CRH (and thus ACTH) is pulsatile
- ACTH has pronounced diurnal pattern with peak in early morning and nadir in late afternoon

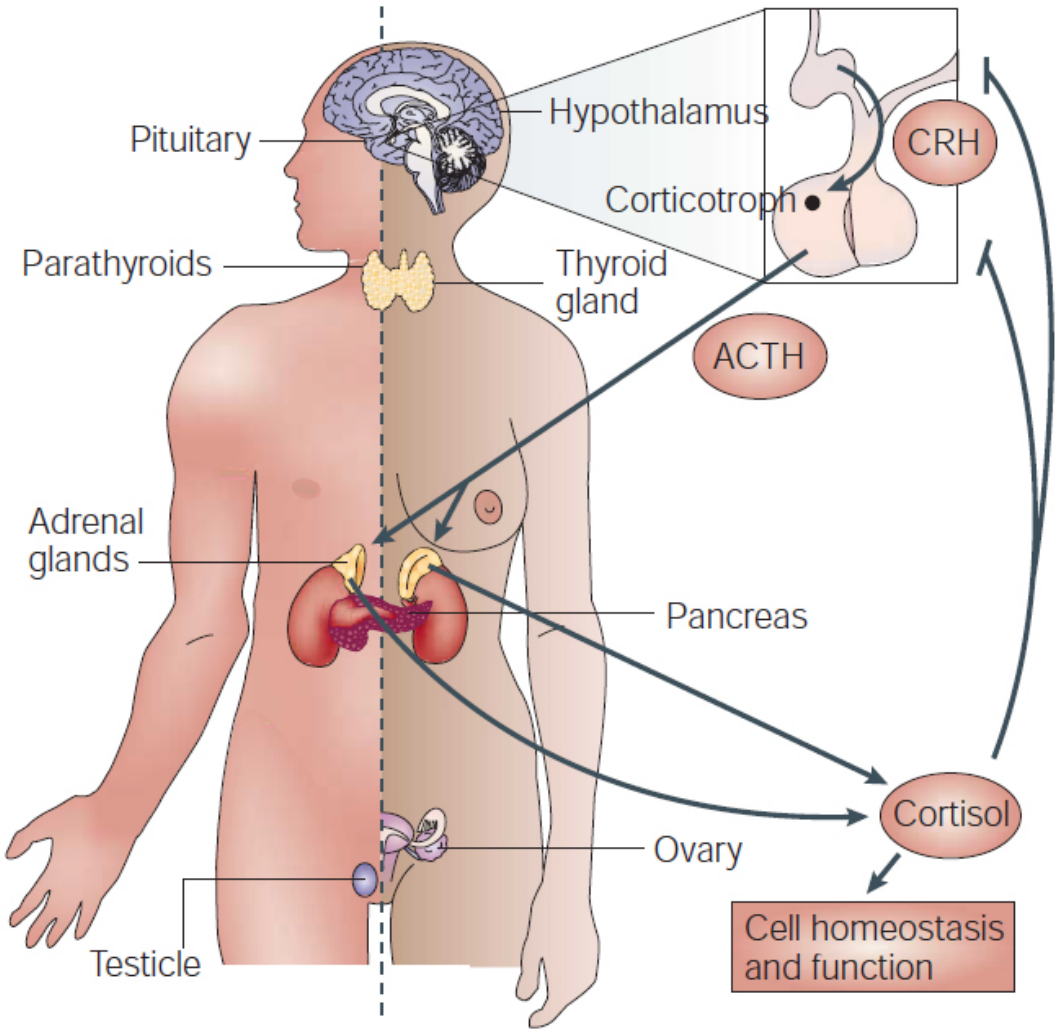


Heaney & Melmed. *Nature Reviews Cancer* **volume 4**, pages 285–295 (2004)

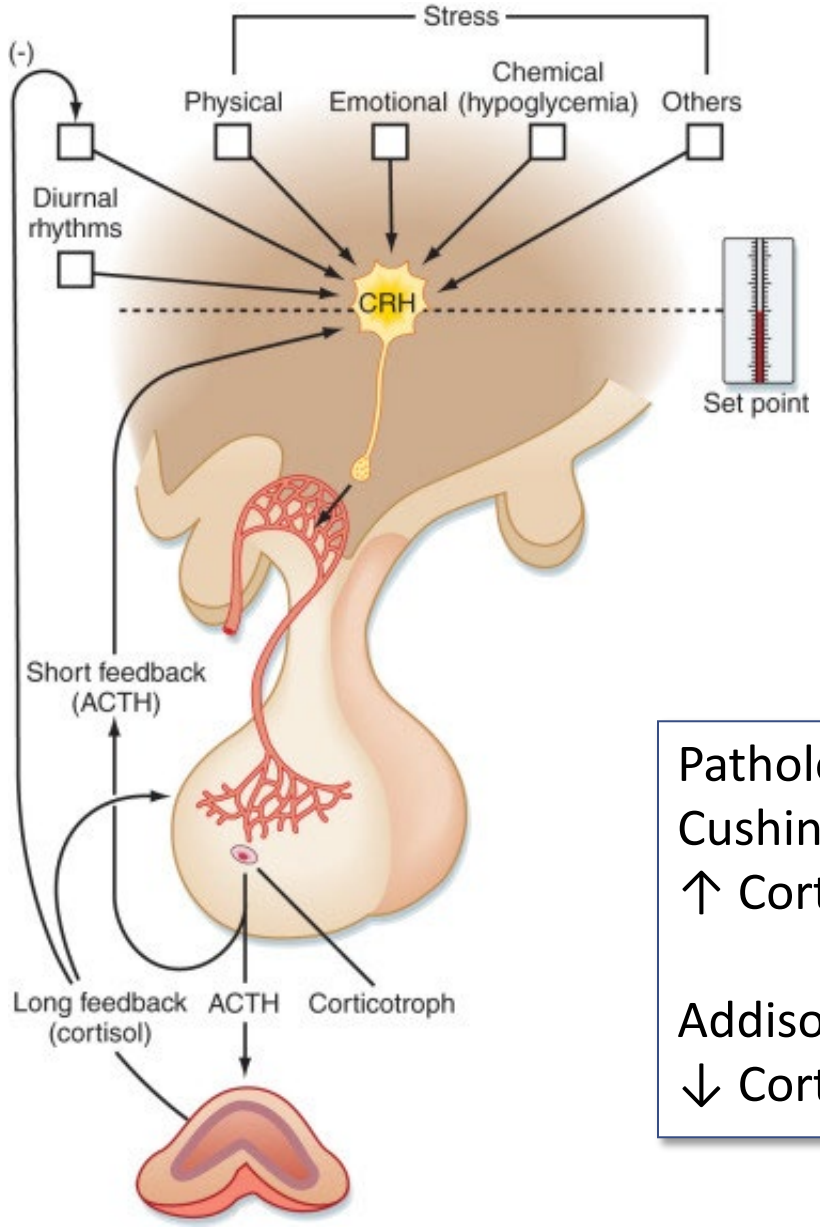
Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 41, 733-752

Stress can alter the set point of the HPA axis

LO3,4



Heaney & Melmed. *Nature Reviews Cancer* **volume 4**, pages 285–295 (2004)



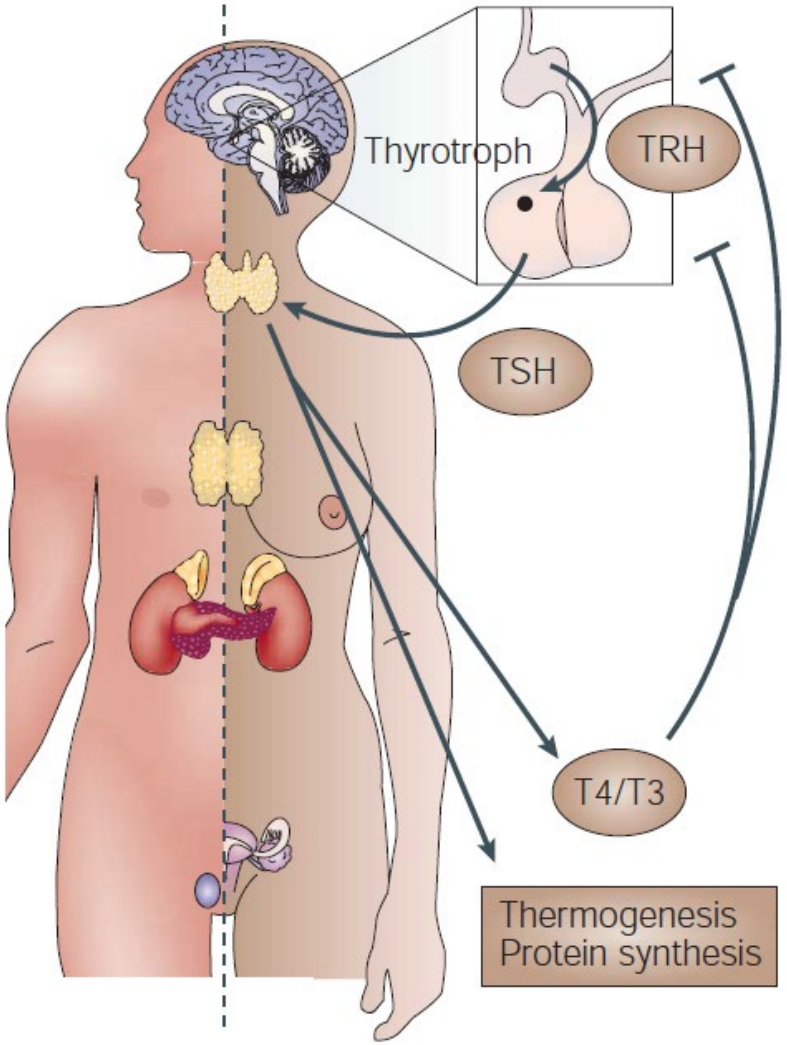
Pathological:
Cushing Syndrome
↑ Cortisol

Addison's disease
↓ Cortisol

Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 41, 733-752

Hypothalamus-Pituitary-Thyroid Axis

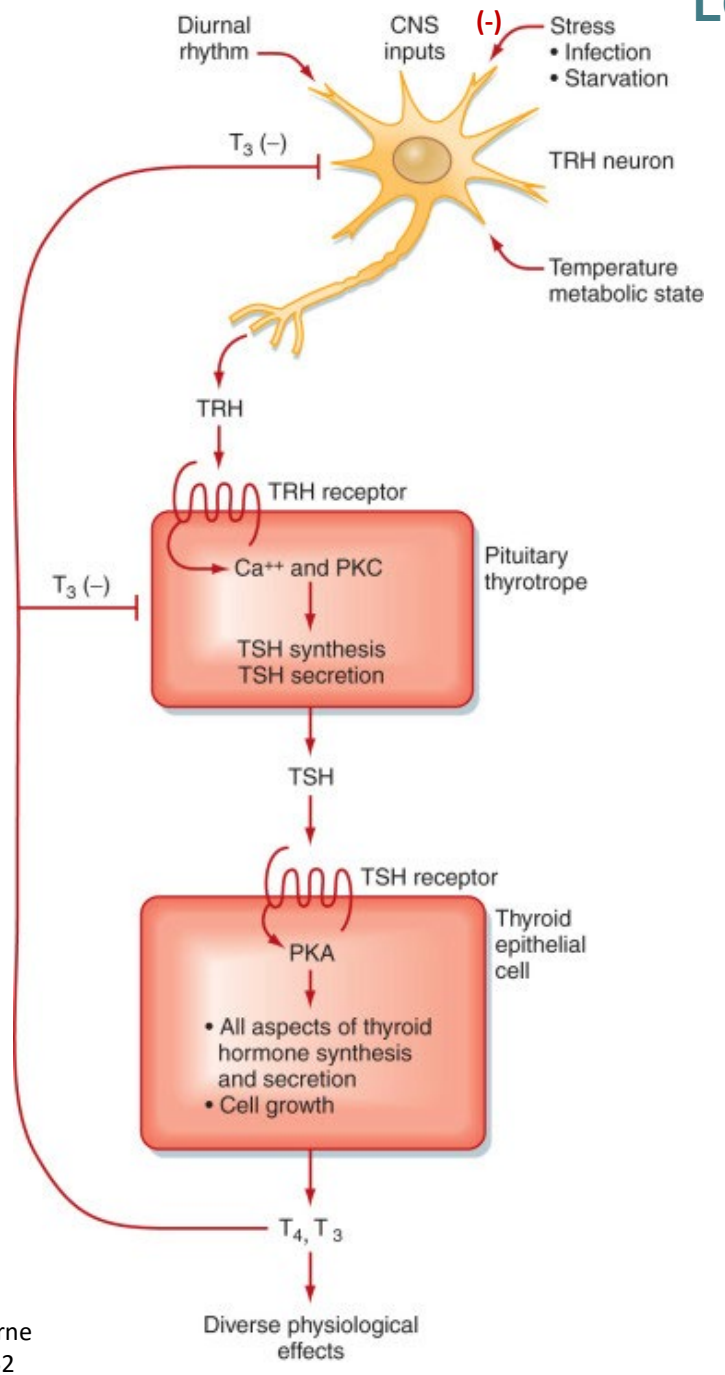
LO3,4



Heaney & Melmed. *Nature Reviews Cancer* **volume 4**, pages 285–295 (2004)

Pathological:
Graves' disease
↑ T4/T3

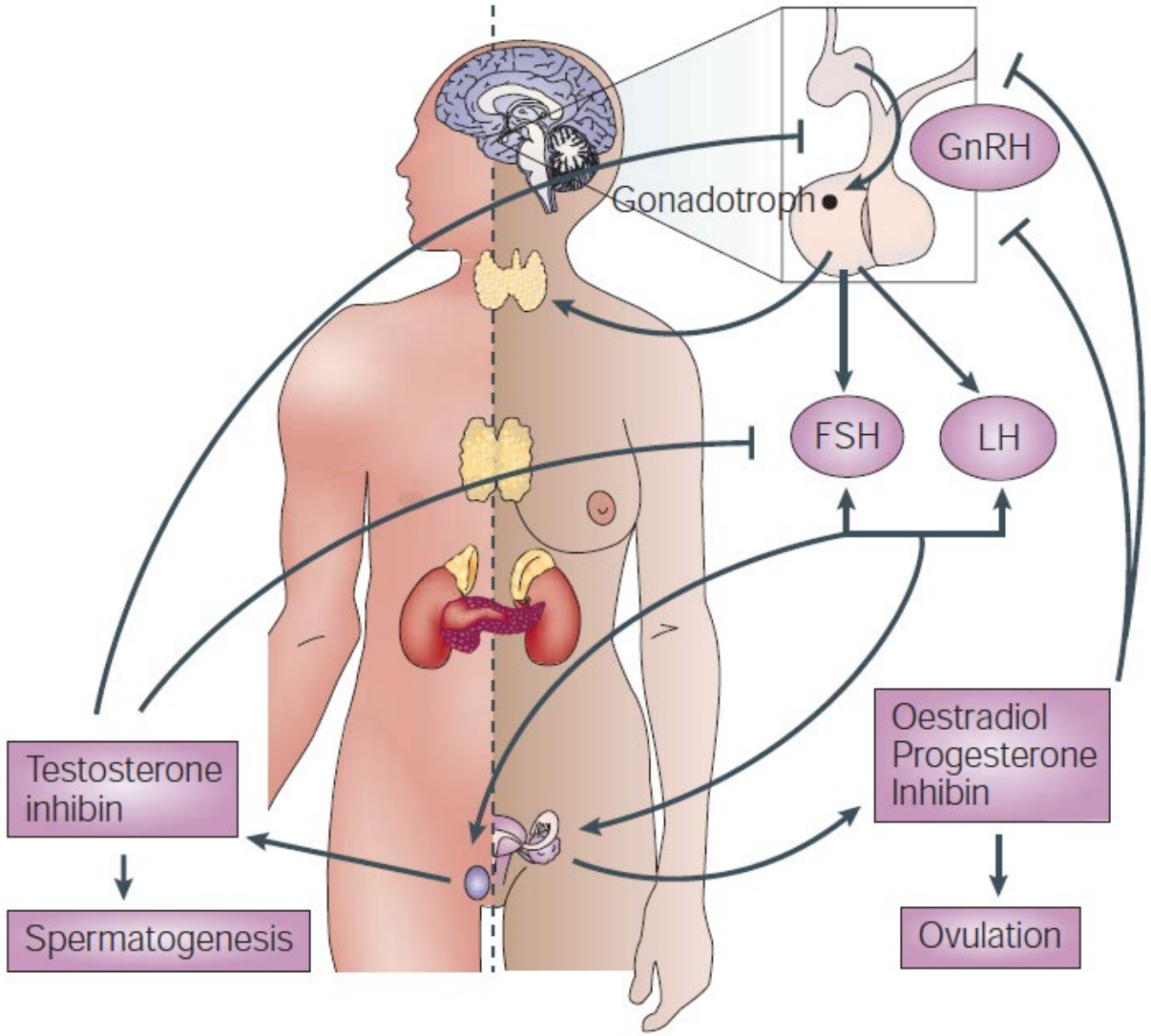
Chronic autoimmune
(Hashimoto's) thyroiditis
↓ T4/T3



Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 41, 733-752

Hypothalamus-Pituitary-Gonad Axis

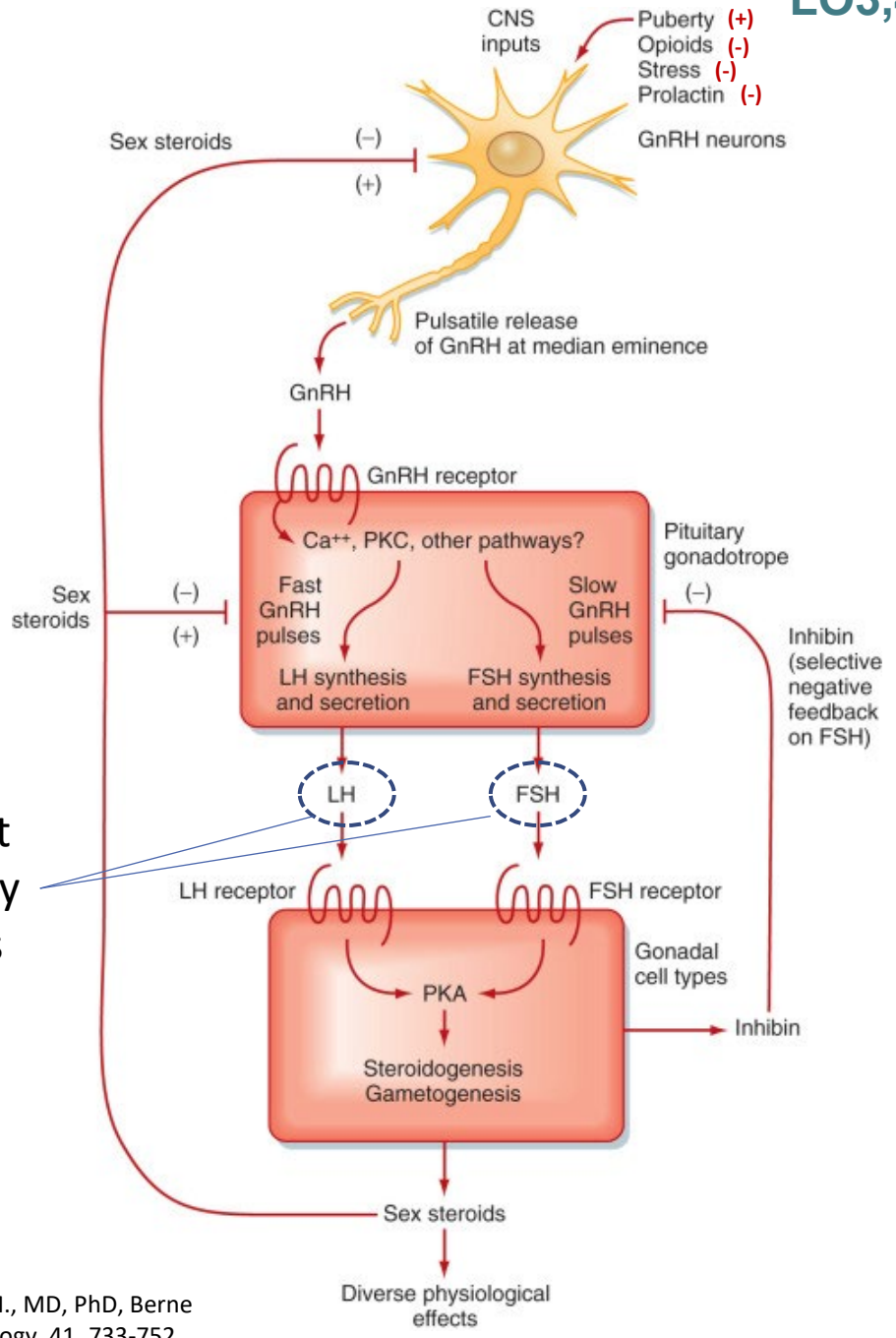
LO3,4



Heaney & Melmed. *Nature Reviews Cancer* **volume 4**, pages 285–295 (2004)

Pathological: Infertility

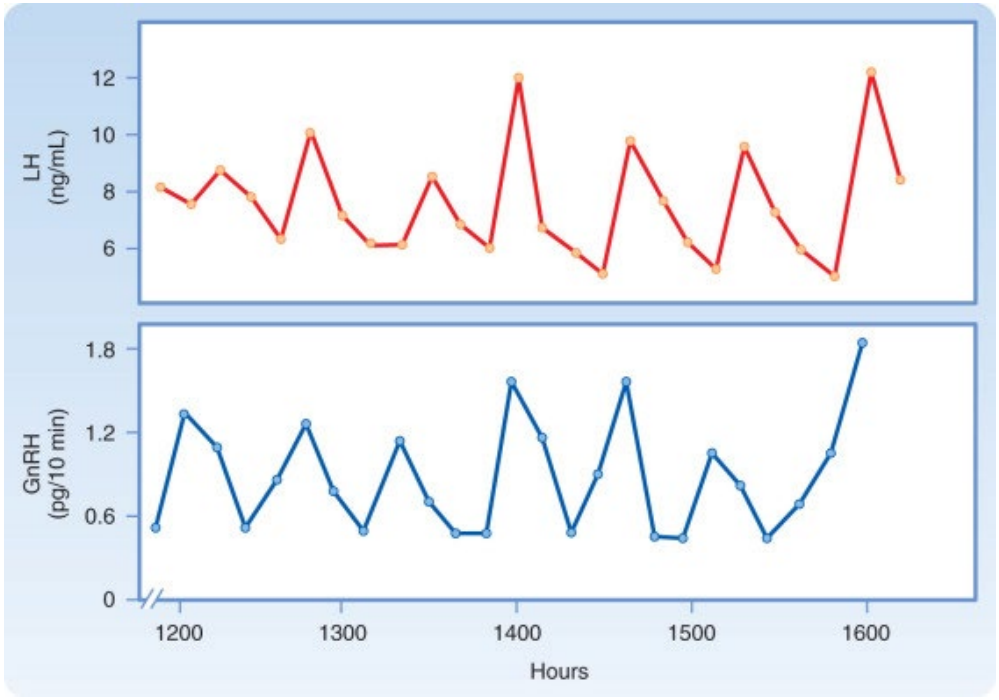
Different secretory granules

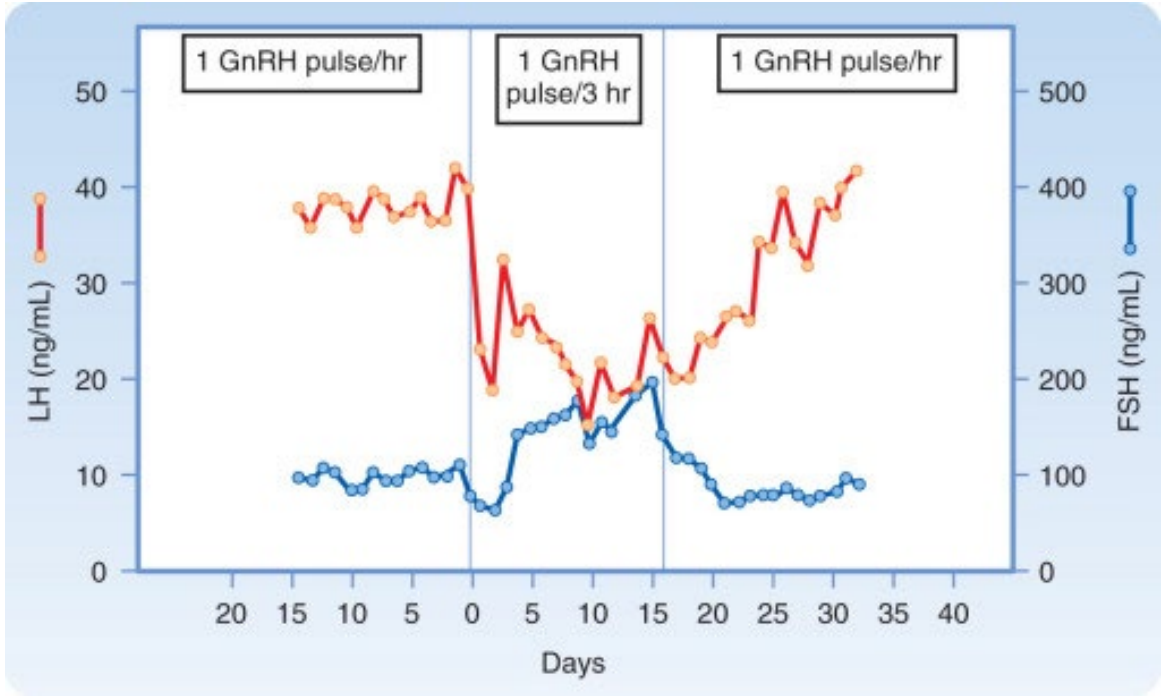


Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 41, 733-752

GnRH has pulsatile release; frequency of pulses dictates release of LH vs FSH

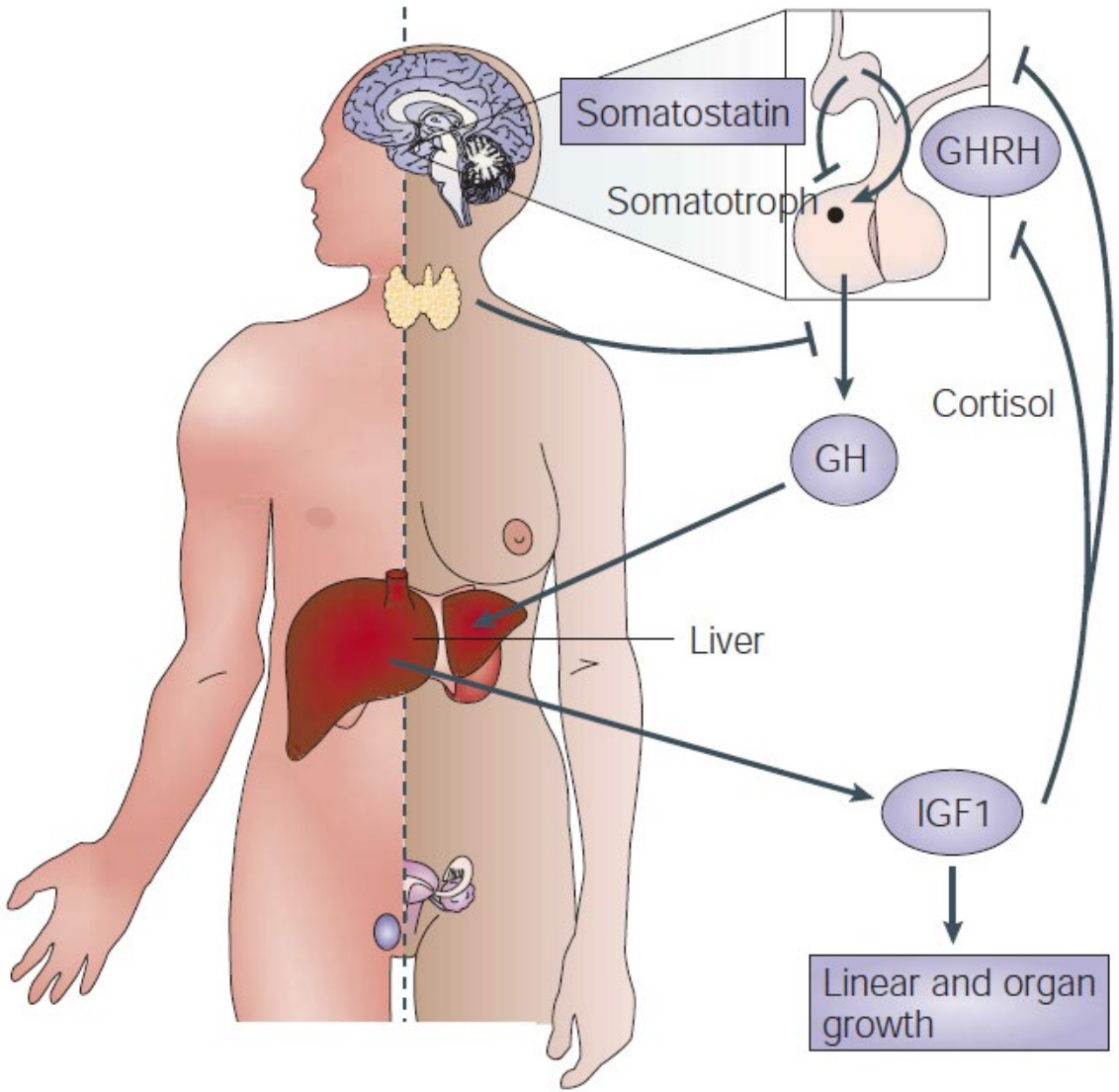
LO3,4





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Hypothalamus-Pituitary-Liver Axis



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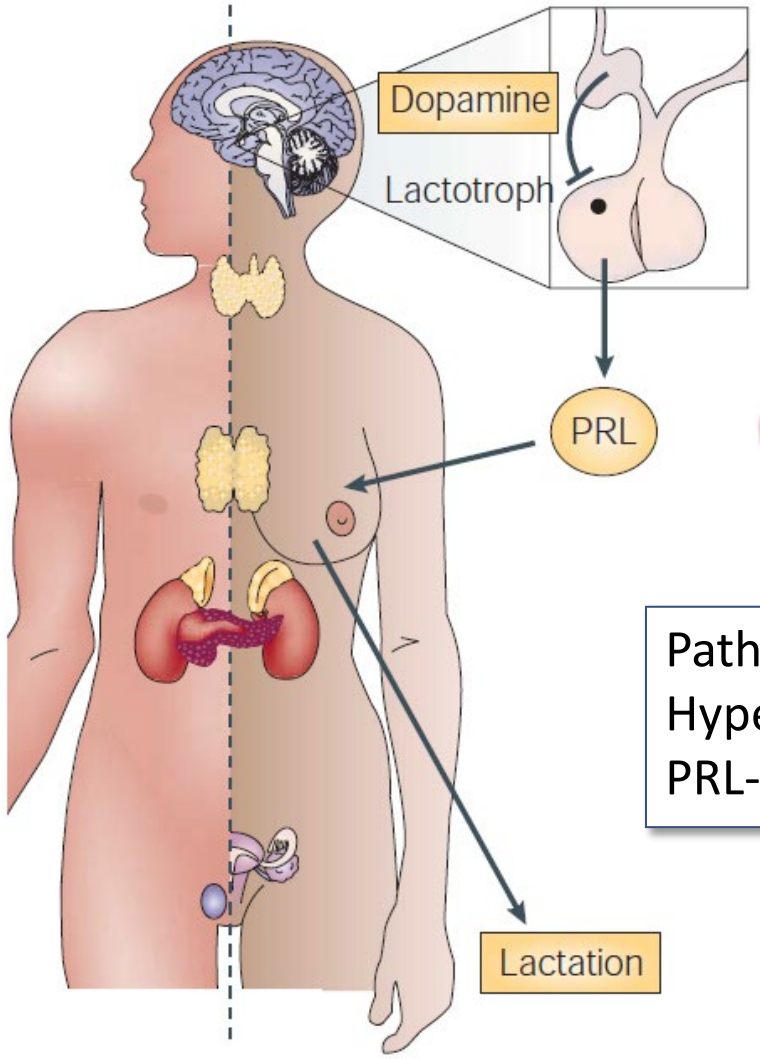
- GH secretion has diurnal rhythm with lowest levels during waking hours
- GHRH promotes GH secretion and somatostatin inhibits
- IGF-1 inhibits through long feedback loop

Pathological:
Laron syndrome
↓ IGF-1

Gigantism; Acromegaly
↑ Growth Hormone

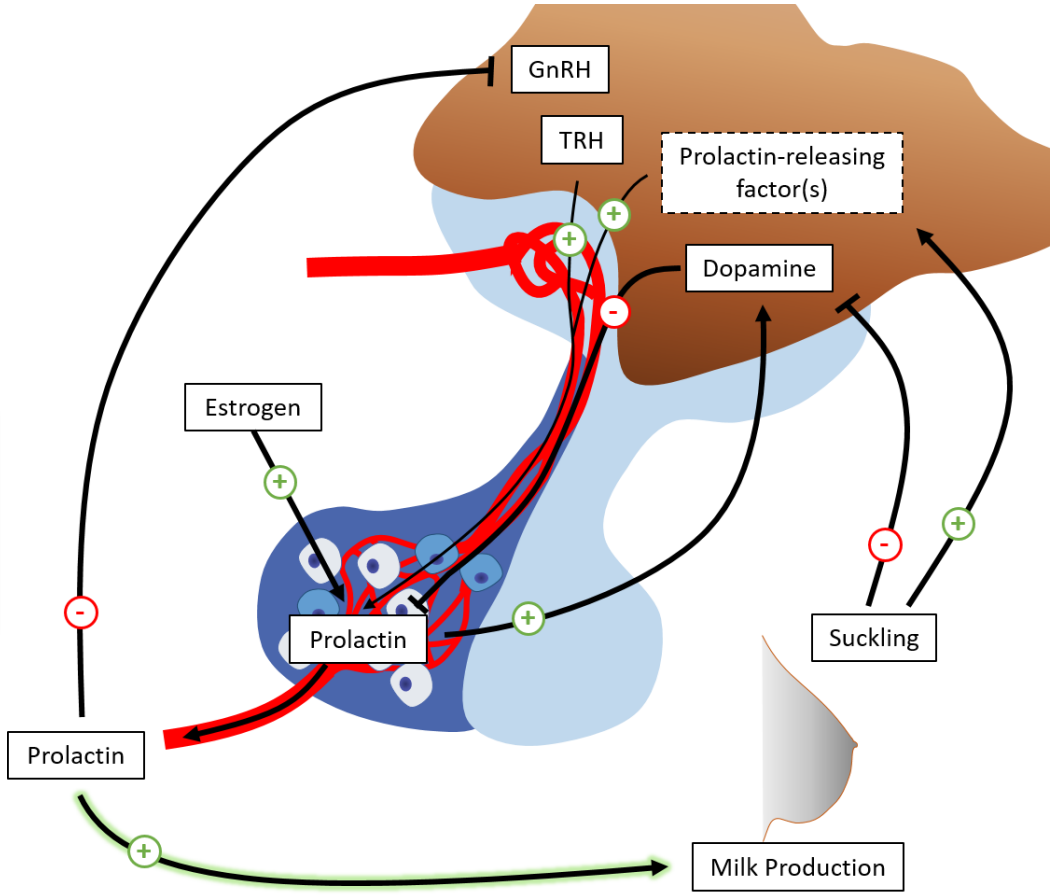
Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 41, 733-752

Hypothalamus-Pituitary-Mammary Gland



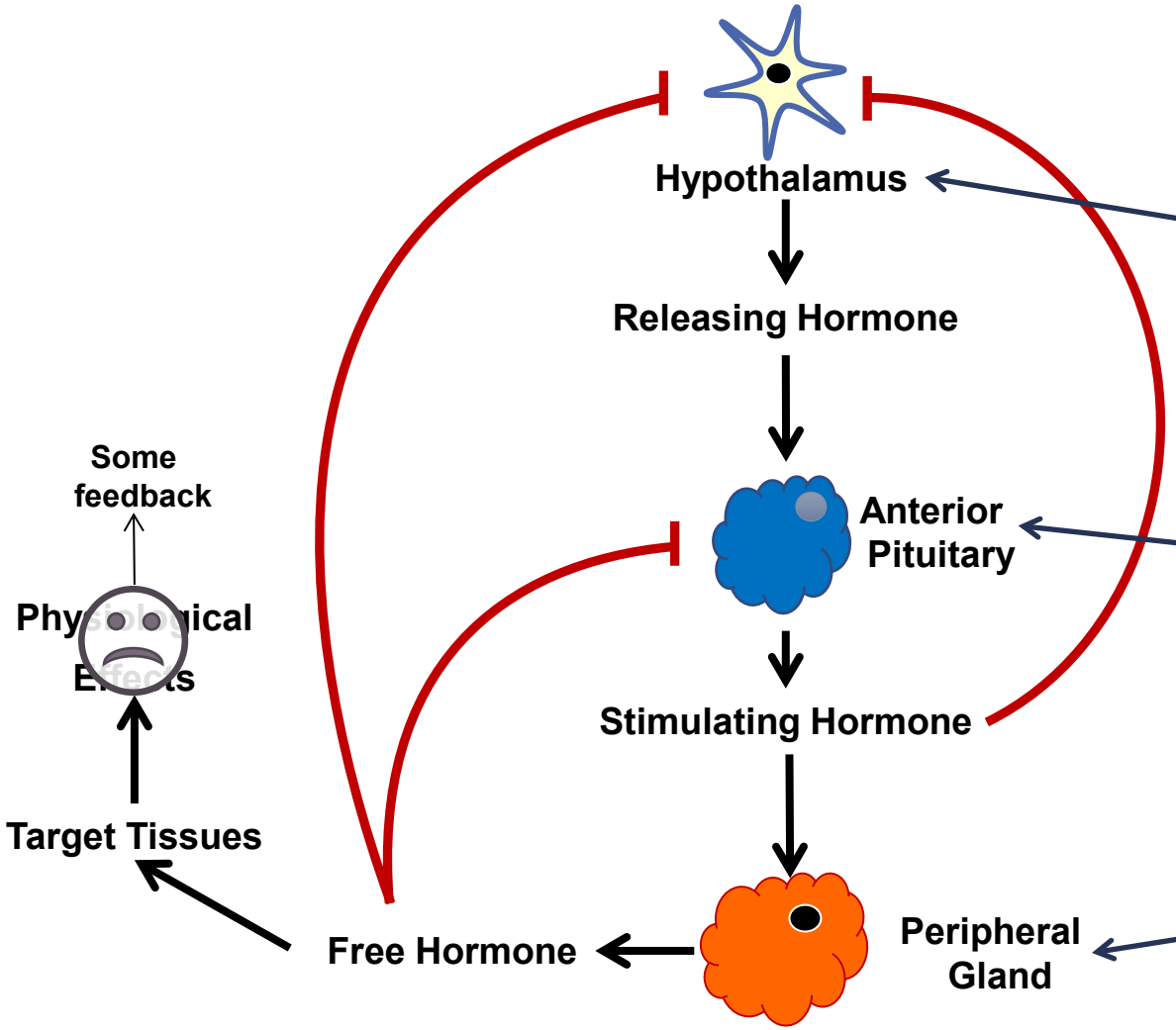
- Prolactin (PRL) acts directly on nonendocrine cells
- Release of PRL normally under tonic inhibition by dopamine

Pathological:
Hyperprolactinemia
PRL-secreting tumors



Heaney & Melmed. *Nature Reviews Cancer* **volume 4**, pages 285–295 (2004)

Endocrine diseases are described clinically as primary, secondary, or tertiary



Tertiary = lesion in hypothalamus

e.g. tertiary hypogonadism

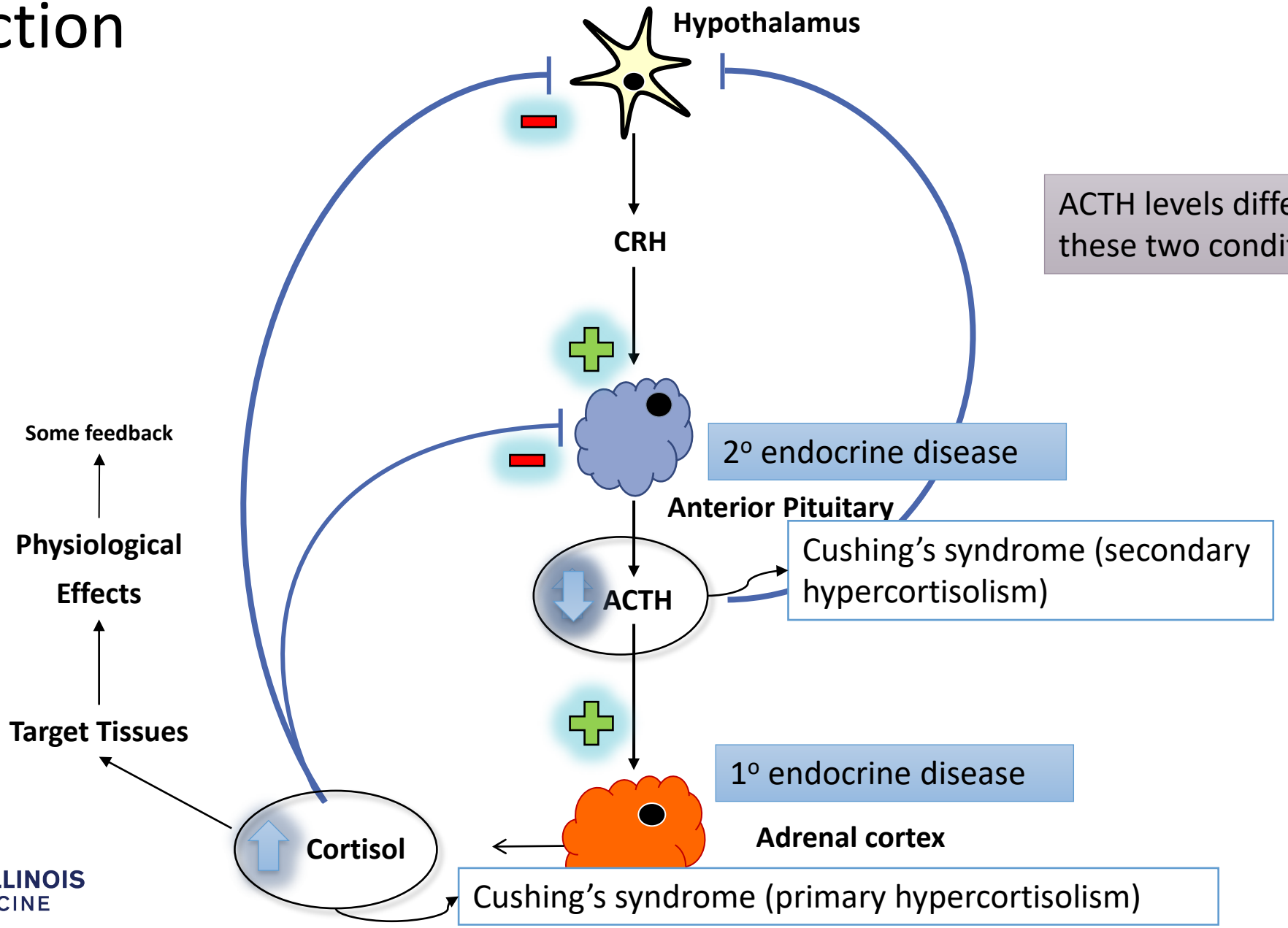
Secondary = lesion in anterior pituitary gland

e.g. secondary hyperthyroidism

Primary = lesion in peripheral endocrine gland

e.g. primary hyperparathyroidism

Cushing's syndrome can result from a 1° or 2° endocrine dysfunction



ACTH levels differentiate these two conditions

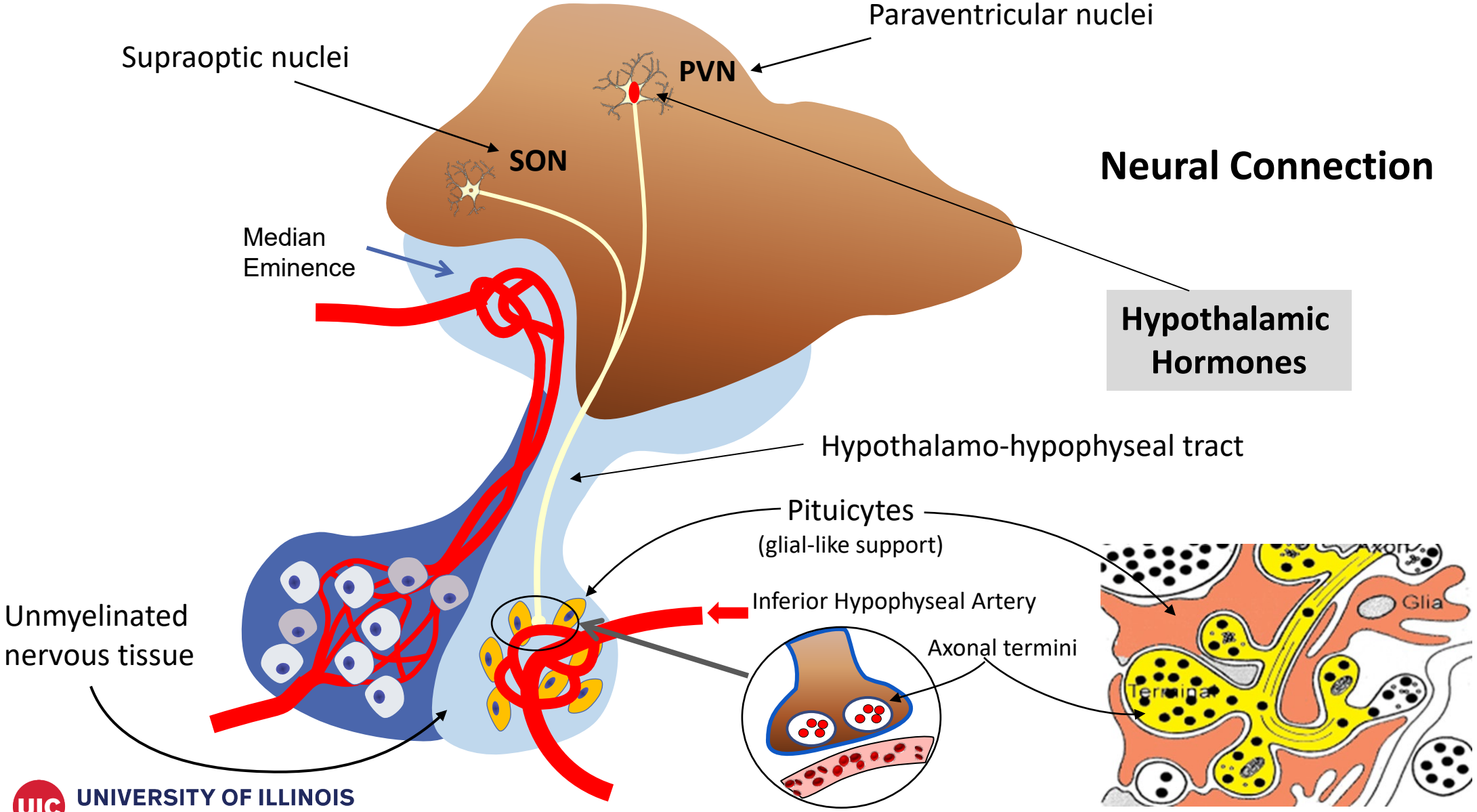
Answer key at end

Try some hypogonadism examples:

Condition	1°	2°	3°
Anabolic steroids provide negative feedback and suppress gonadotropin secretion			
Adenoma functionally disrupts gonadotroph cells			
Low testosterone; high FSH/LH; high GnRH			
Low testosterone; low FSH/LH; low GnRH			
Hyperprolactinemia			
Low testosterone; low FSH/LH; high GnRH			
Kallman syndrome (caused by congenital defects in GnRH-producing neurons)			
Leydig cell hypoplasia directly affecting testes			

Neurohypophysis

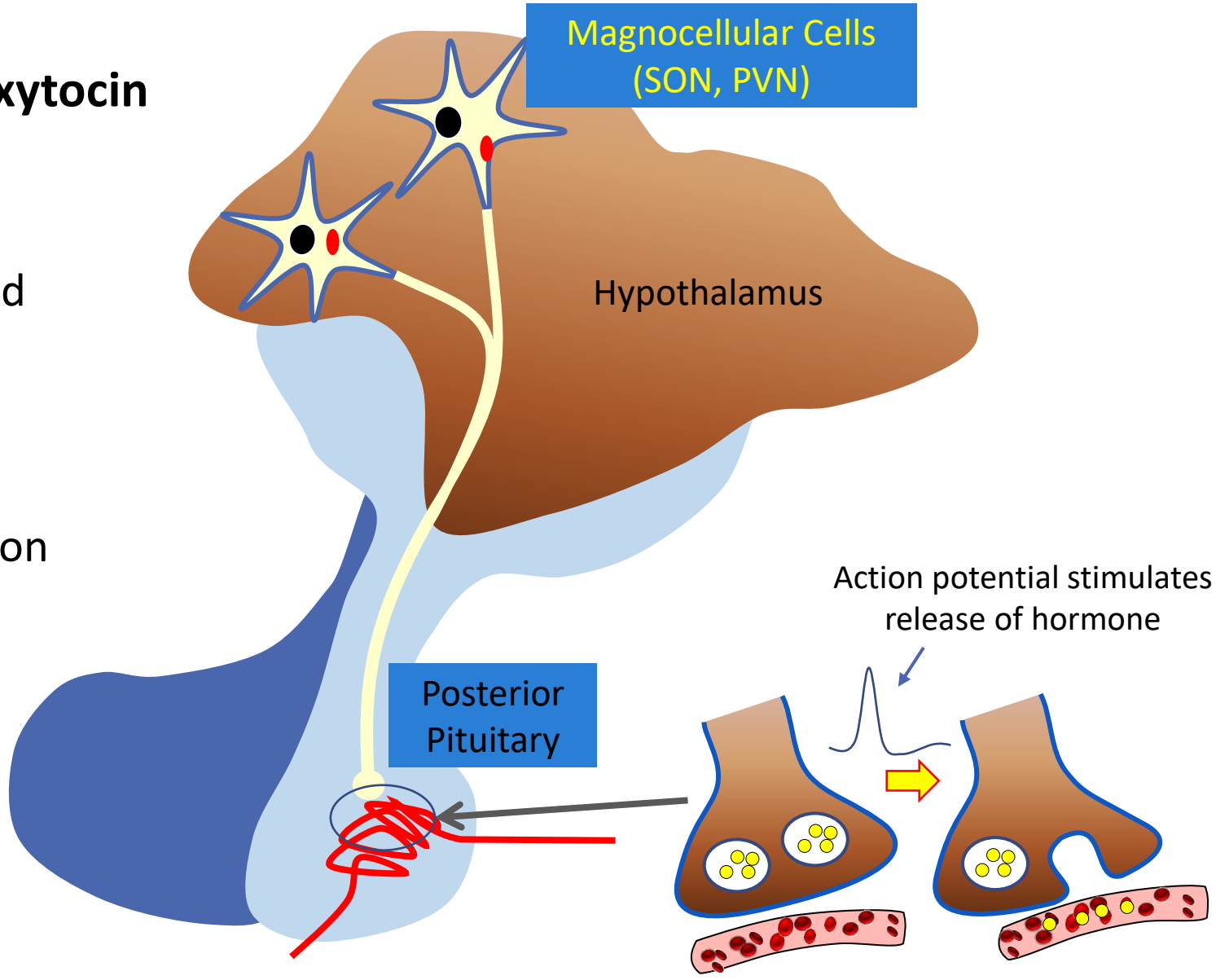
Hypothalamus-Neurohypophysis Connection



Posterior pituitary hormone synthesis and release

Hormones: **ADH/Vasopressin** and **Oxytocin**

- Peptide hormones
- Synthesized in magnocellular cells as preprohormones, which are processed during transport to storage site
- Stored in axon terminal (in posterior pituitary) in secretory granules
- Released via exocytosis following action potential



Primary functions of posterior pituitary hormones oxytocin and ADH/vasopressin

Oxytocin:

- Promotes contraction of uterine smooth muscle during **childbirth**
- Promotes contraction of myoepithelial cells in the breast to initiate **milk ejection**
- Acts within CNS structures to reduce stress response and promote nurturing and bonding behaviors
- Has other effects on behavior which are still being characterized

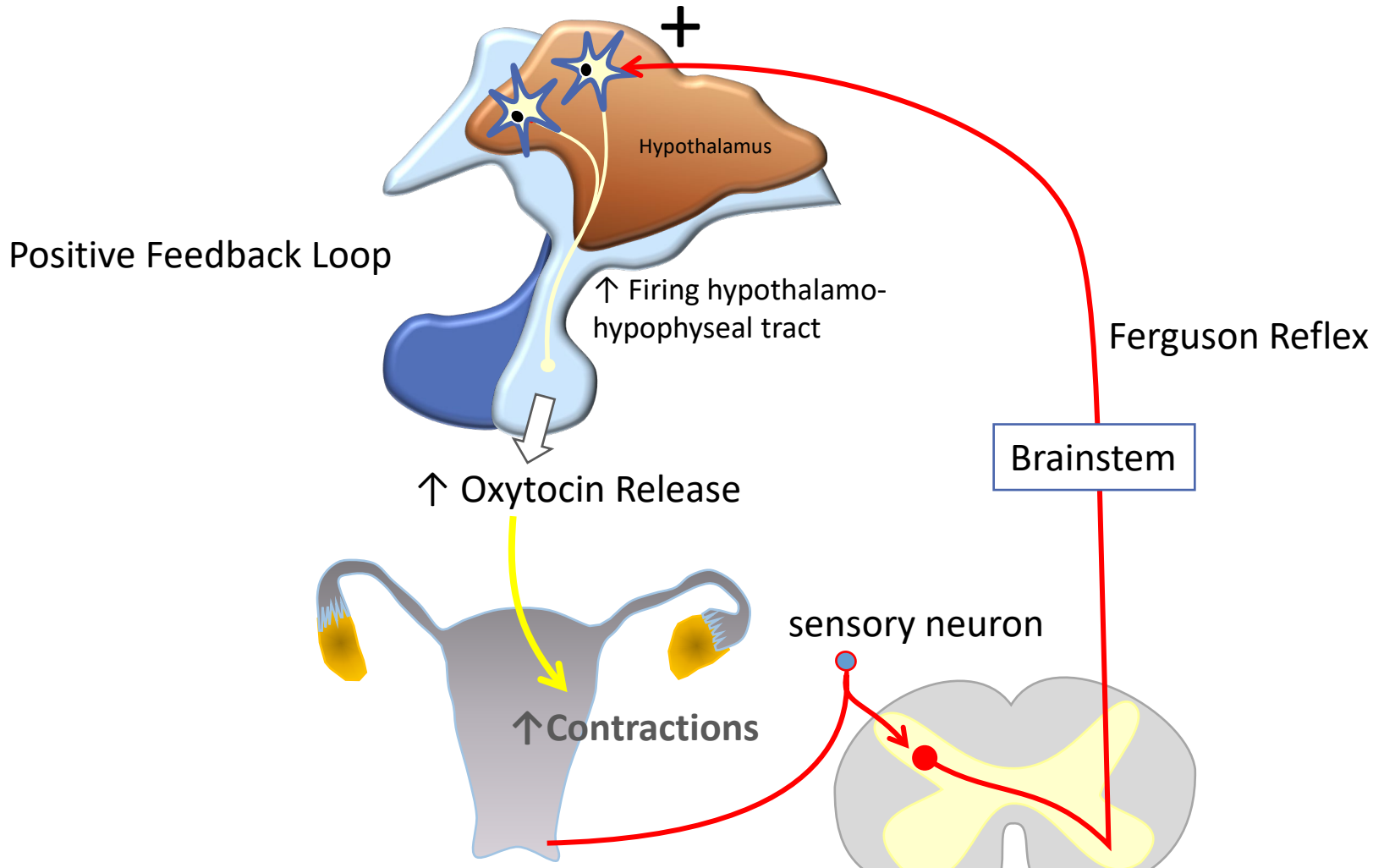
Oxytocin receptor is a GPCR linked to $G_{\alpha q}$
Phospholipase C → IP_3 → calcium release → contraction

Vasopressin (ADH):

- Promotes contraction of vascular smooth muscle cells, constricting arterioles and **raising blood pressure**
- Promotes **renal water reabsorption** through insertion of aquaporins
- Stimulates vascular endothelial cells to release von Willebrand factor to promote clotting

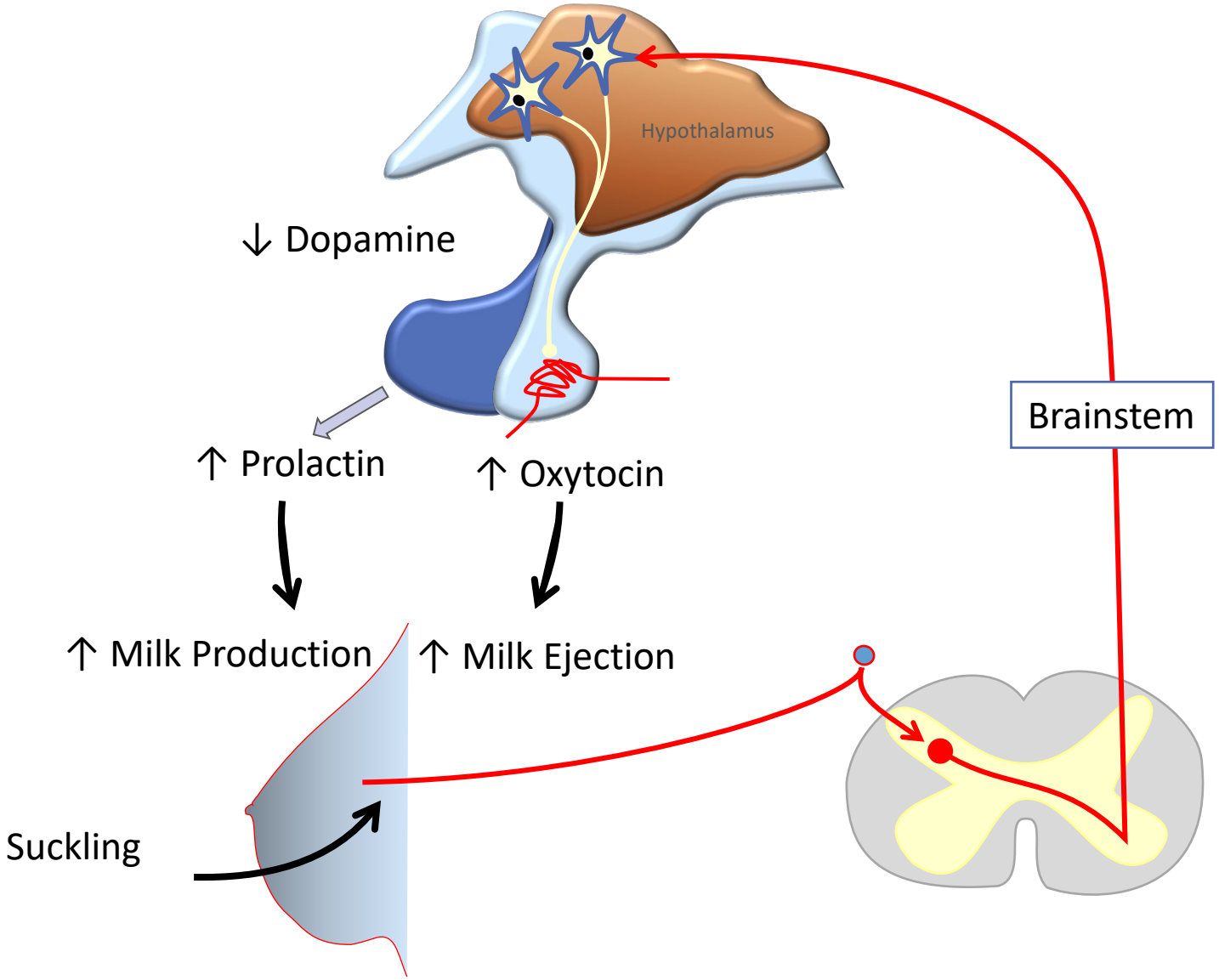
ADH receptors are GPCRs:
- V_1 linked to $G_{\alpha q}$
- V_2 linked to $G_{\alpha s}$

Regulation of oxytocin secretion by myometrial contractions
- a positive feedback loop



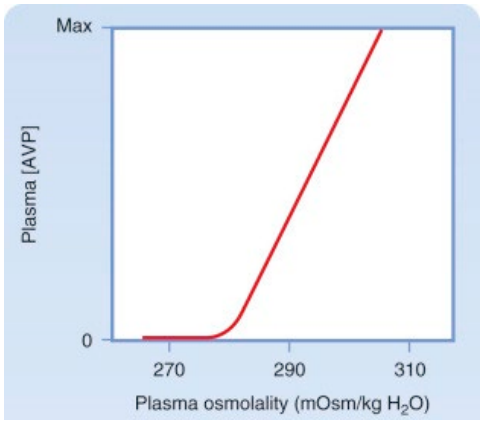
Labor ➡ ↑ Myometrial Contractions ➡ ↑ Stretching of vagina and cervix

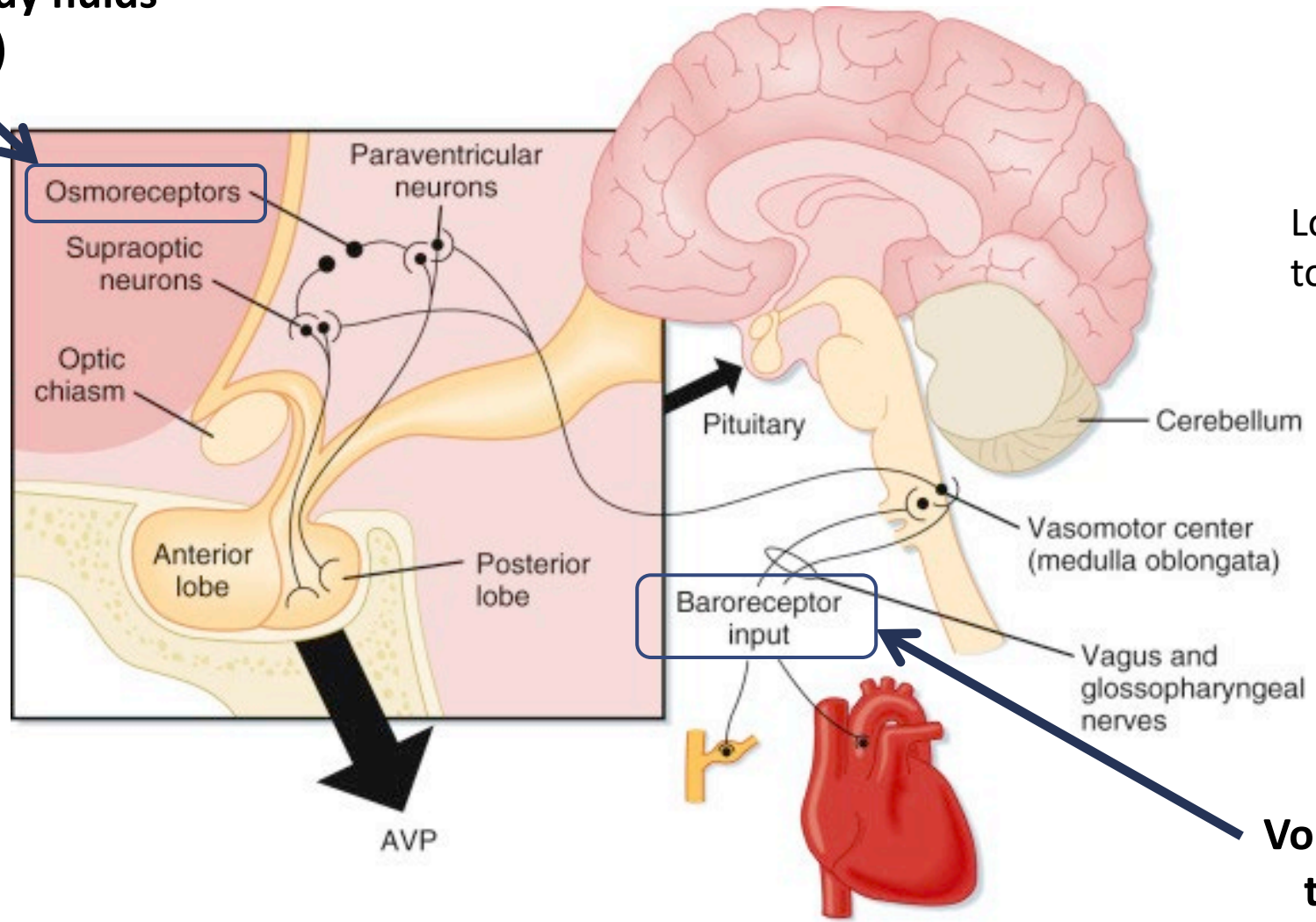
Oxytocin is also secreted in response to suckling



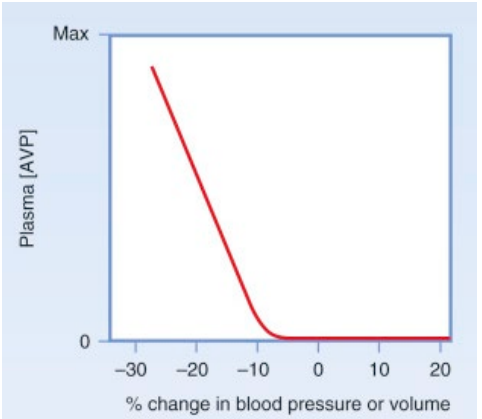
Regulators of ADH/vasopressin (AVP) secretion

Osmolality of body fluids (osmotic)





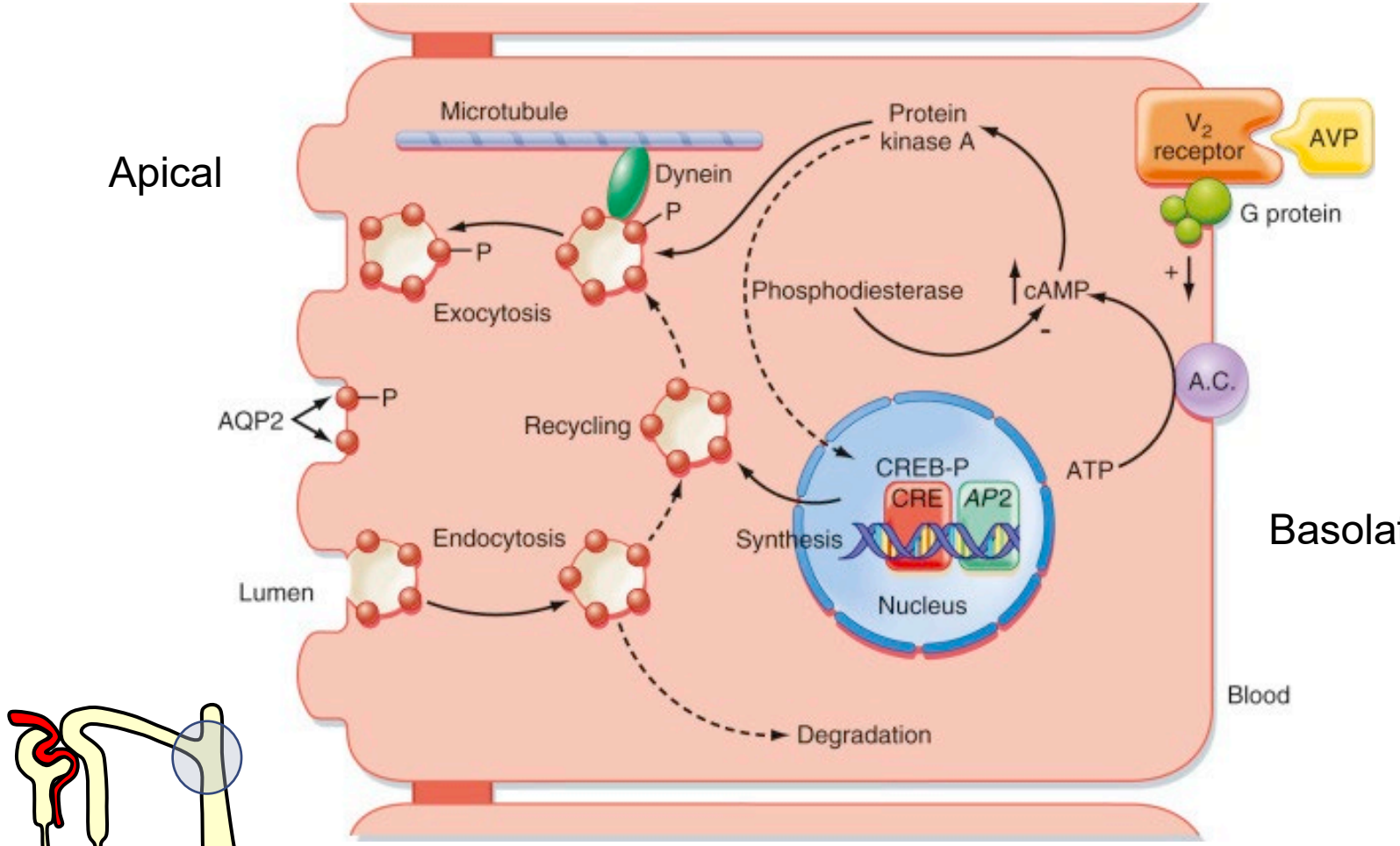
Low volume ↑ sensitivity to osmolality changes

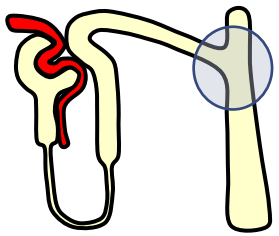


Volume and pressure of the vascular system (hemodynamic)

Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 35, 623-646

ADH enhances water absorption from tubular fluid and stimulates Na^+ reabsorption





↑ urine osmolality, ↓ serum osmolality

Pathological:

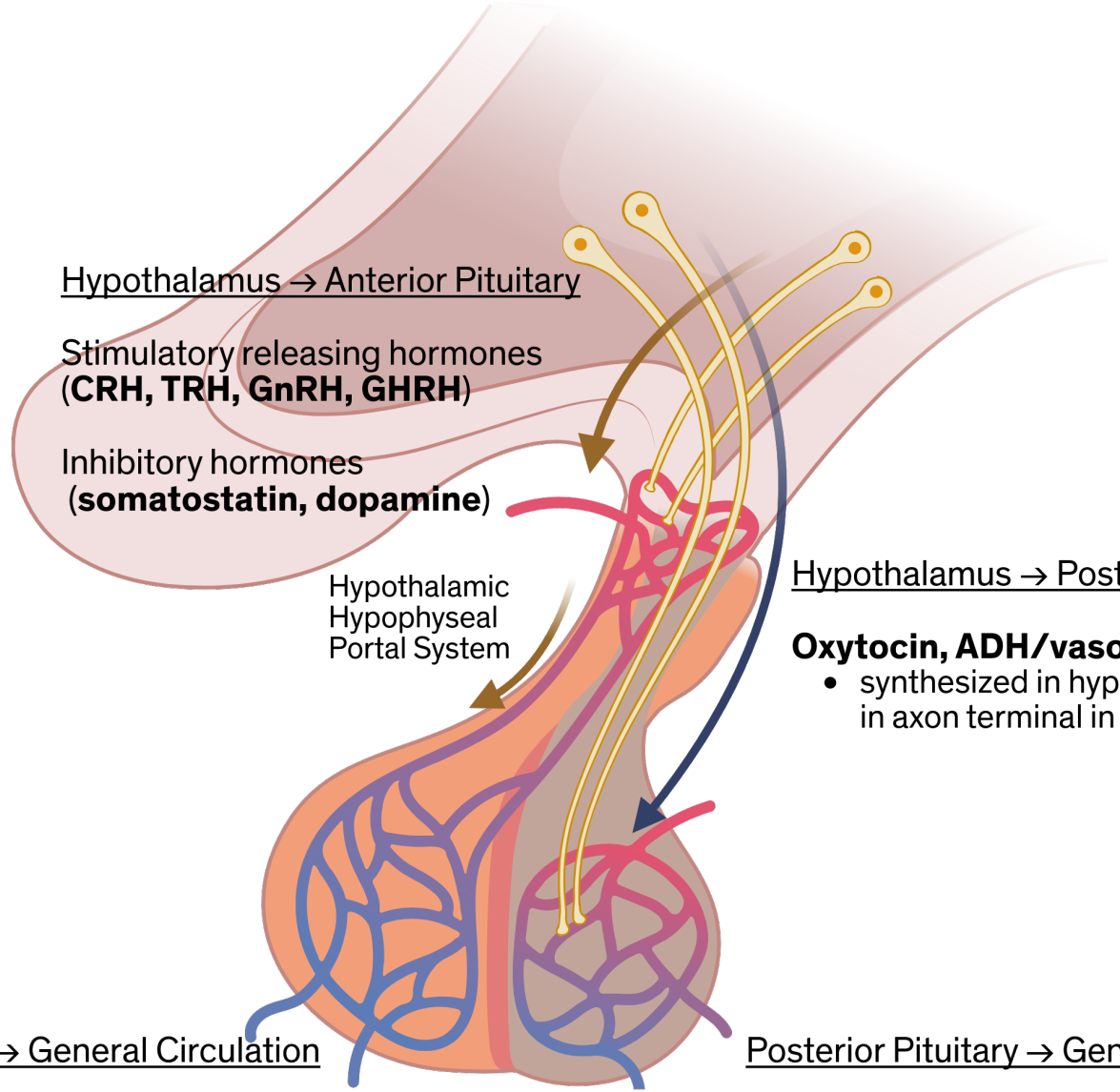
Central diabetes insipidus
- Head trauma, brain cancer, infection
- Inherited mutations in AVP

Syndrome of inappropriate ADH secretion (SIADH)
- Brain infections and cancer, drugs, pulmonary diseases, lung cancer

Nephrogenic diabetes insipidus
- Decreased expression of AQP2
- Inherited mutations in V_2 receptor or AQP2 genes
- Resistance to ADH caused by chronic lithium ingestion

Koeppen, Bruce M., MD, PhD, Berne and Levy Physiology, 35, 623-646

Summary of hypothalamus - pituitary connections



Hypothalamus → Anterior Pituitary

Stimulatory releasing hormones
(**CRH, TRH, GnRH, GHRH**)

Inhibitory hormones
(**somatostatin, dopamine**)

Hypothalamic
Hypophyseal
Portal System

Hypothalamus → Posterior Pituitary

Oxytocin, ADH/vasopressin

- synthesized in hypothalamus, stored in axon terminal in posterior pituitary

Anterior Pituitary → General Circulation

ACTH, TSH, FSH, LH, GH, PRL

Posterior Pituitary → General Circulation

release of hormone upon stimulus

Summary- Anterior Pituitary

- Adenohypophysis contains 5 cell types that produce 6 hormones: ACTH, TSH, FSH, LH, GH, and PRL
- In endocrine axis regulation, long negative feedback to anterior pituitary and hypothalamus is provided by levels of free hormone and short negative feedback to hypothalamus by levels of stimulating hormone
- Hypothalamus-pituitary-adrenal axis: CRH → ACTH → Cortisol
  - Stimulated by stress (many forms)
  - Pronounced diurnal pattern with peak in early morning
- Hypothalamus-pituitary-thyroid axis: TRH → TSH → Thyroid Hormone
  - Downregulated by stress
  - Negative feedback comes primarily from biologically active T_3
- Hypothalamus-pituitary-gonad axis: GnRH → FSH and LH → Testosterone, Estrogen, Progesterone, Inhibin
  - Pulsatile release of GnRH
  - Sex steroids provide negative feedback
- Hypothalamus-pituitary-liver axis: GHRH → GH → IGF-1
  - Both GH and IGF-1 have systemic physiological effects
  - Somatostatin also inhibits somatotrophs
- Hypothalamus-pituitary-mammary gland axis
  - PRL release from lactotrophs normally inhibited by dopamine from hypothalamus
- Endocrine diseases classified as 1°, 2°, or 3° based on level of lesion

Summary- Neurohypophysis (Posterior Pituitary)

- Two hormones produced: ADH and oxytocin
- Oxytocin regulated by myometrial contractions (positive feedback loop) and suckling during breastfeeding
  - GPCR → IP_3 → myoepithelial contraction
- ADH regulated by changes in plasma osmolality and hemodynamics
  - Enhances water absorption from tubular fluid in distal tubule and collecting duct
  - Involved in several diseases

Answers from slide 21:

Condition	1°	2°	3°
Anabolic steroids provide negative feedback and suppress gonadotropin secretion		X	
Adenoma functionally disrupts gonadotroph cells		X	
Low testosterone; high FSH/LH; high GnRH	X		
Low testosterone; low FSH/LH; low GnRH			X
Hyperprolactinemia			X
Low testosterone; low FSH/LH; high GnRH		X	
Kallman syndrome (caused by congenital defects in GnRH-producing neurons)			X
Leydig cell hypoplasia directly affecting testes	X		